Organic Chemistry Booklet

Midterm · Lectures 1 – 9

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Each lecture is split into topics with its study guide questions (multiple choice). The answer key is at the end of each lecture.

1 Structural Isomerism

In this lecture

1. Isomers & Structural Isomers 13 questions
2. Stereoisomers & Geometric Isomerism 3 questions
3. Chirality & Chiral Carbon 6 questions
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1.1 Isomers & Structural Isomers

Q1 What are isomers?

A. Compounds with the same structural formula

B. Compounds with the same molecular formula

C. Compounds with the same physical properties

D. Compounds with different molecular weights

Q2 Structural (constitutional) isomers have:

A. The same molecular formula but atoms bonded in a different order

B. The same order of bonding but a different arrangement in space

C. Different molecular formulas

D. Identical physical and chemical properties

Q3 Dimethyl ether (CH_3OCH_3) and ethanol (CH_3CH_2OH) have the same physical properties.

True

False

Q4 Butane ($CH_3CH_2CH_2CH_3$) and methylpropane (isobutane, $(CH_3)_2CHCH_3$) are structural isomers of C_4H_{10} with different boiling points.

True

False

Q5 Which of the following is **NOT** a structural isomer of C_4H_8?

A. Butane ($CH_3CH_2CH_2CH_3$)

B. 1-Butene ($CH_2{=}CHCH_2CH_3$)

C. Methylcyclopropane

D. Cyclobutane

Q6 Which of the following is **NOT** an isomer of $C_4H_{10}O$?

A. Diethyl ether ($CH_3CH_2OCH_2CH_3$)

B. 1-Propanol

C. 2-Methyl-2-propanol

D. 1-Butanol

Q7 Which of the following is **NOT** a structural isomer of C_6H_{14}?

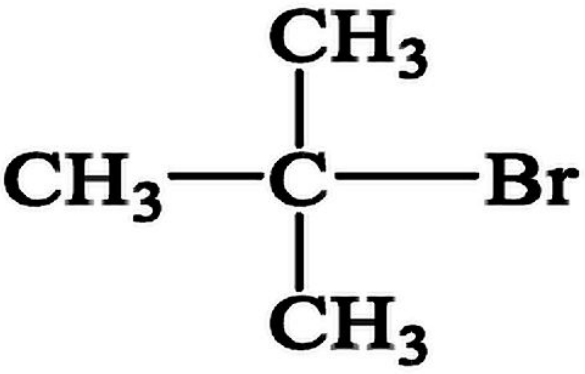
A. 2,2-Dimethylbutane

B. 2-Methylpentane

C. Hexane

D. Cyclohexane

Q8 Which compound is a structural isomer of this compound, $(CH_3)_3CBr$ (2-bromo-2-methylpropane)?



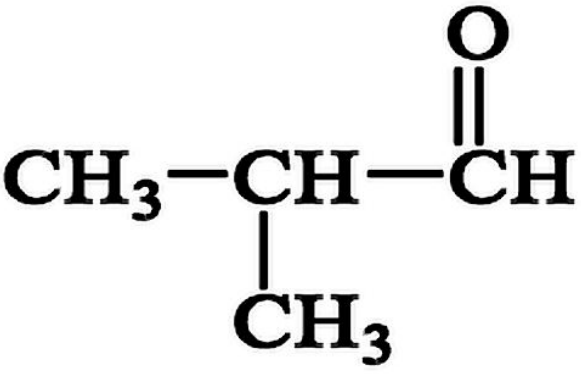
A. $(CH_3)_3CCl$

B. CH_3CH_2Br

C. $CH_3C(CH_3)_2Br$

D. $CH_3CH(CH_3)CH_2Br$

Q9 Which compound is a structural isomer of this compound, 2-methylpropanal (C_4H_8O)?



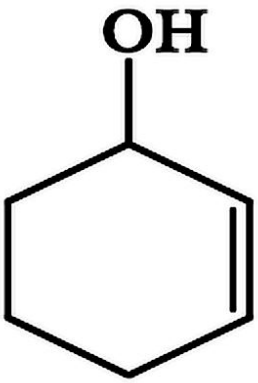
A. Butanone ($CH_3COCH_2CH_3$)

B. 2-Methylpropan-1-ol ($(CH_3)_2CHCH_2OH$)

C. 2-Methylpropanoic acid ($(CH_3)_2CHCOOH$)

D. Propanal (CH_3CH_2CHO)

Q10 Which compound is a structural isomer of this compound, cyclohex-2-en-1-ol ($C_6H_{10}O$)?



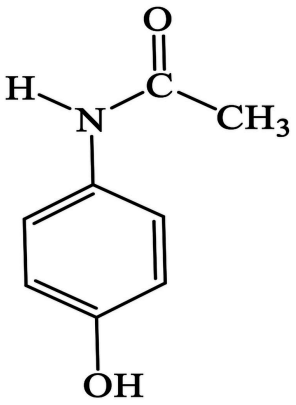
A. Phenol

B. Cyclohexane-1,2-diol

C. Cyclohexanol

D. Cyclohex-3-en-1-ol

Q11 This compound ($C_8H_9NO_2$) has the OH at the para position (opposite the NH group). Which is a structural isomer of it?



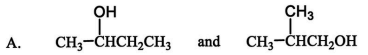
A. The same compound with the OH at the meta position

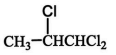
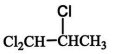
B. The same compound with two OH groups

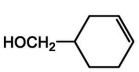
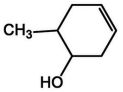
C. The same compound without the OH group

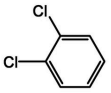
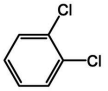
D. The same compound drawn upside down

Q12 Which of the following pairs represent structural isomers?

A.  and

B.  and 

C.  and 

D.  and 

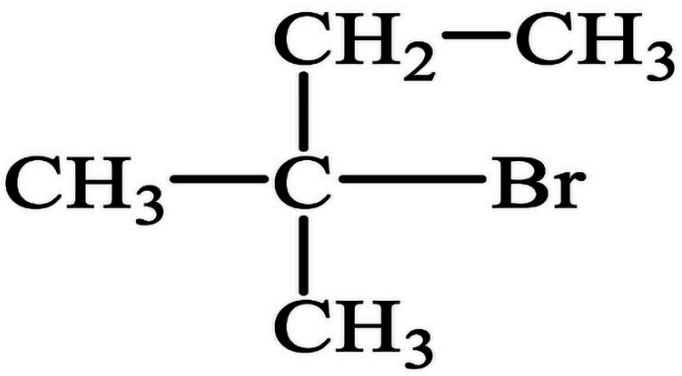
A. A only

B. A, B, C and D

C. A and C

D. B and D

Q13 Which is a structural isomer of 2-bromo-2-methylbutane, shown below?



A. 1-Bromopentane

B. 1-Bromopropane

C. 2-Bromo-2-methylpropane

D. 2-Bromobutane

1.2 Stereoisomers & Geometric Isomerism

Q14 What type of isomerism exists in compounds containing carbon-carbon double bonds and cyclic compounds?

A. Meso isomerism

B. Structural isomerism

C. Geometric isomerism

D. Functional isomerism

Q15 What is the major difference between constitutional isomers and stereoisomers?

A. Constitutional isomers are bonded in different orders; stereoisomers are bonded in the same order but oriented differently in space

B. There is no difference

C. Stereoisomers have different molecular formulas

D. Stereoisomers are bonded in different orders; constitutional isomers differ only in space

Q16 Which of the following is **NOT** a type of stereoisomer?

A. Diastereomers

B. Constitutional isomers

C. Enantiomers

D. Geometric isomers

1.3 Chirality & Chiral Carbon

Q17 Which of the following is a **chiral** object?

A. A hand

B. A cup

C. A conical flask

D. A ball

Q18 An achiral object can be superposed on its mirror image.

True

False

Q19 A chiral carbon is an sp^3-hybridized carbon with:

A. Two identical groups attached

B. Four identical groups attached

C. A double bond

D. Four different atoms or groups attached

Q20 Another name for a chiral carbon is:

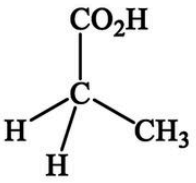
A. Meso center

B. Achiral carbon

C. Stereocenter

D. sp^2 carbon

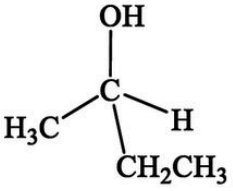
Q21 Which of the following molecules contains a chiral carbon?



a.
Propanoic acid



b.
Cyclohexanol



c.
2-Butanol

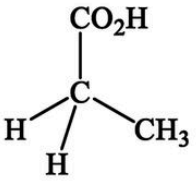
A. a and c

B. c. 2-Butanol

C. b. Cyclohexanol

D. a. Propanoic acid

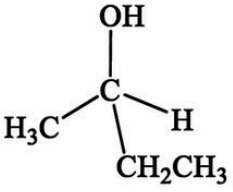
Q22 Which of the following compounds can exist as a pair of optically active compounds (enantiomers or mirror images)?



a.
Propanoic acid



b.
Cyclohexanol



c.
2-Butanol

A. a. Propanoic acid

B. c. 2-Butanol

C. b. Cyclohexanol

D. a and c

1.4 Optical Activity & Enantiomers

Q23 A dextrorotatory enantiomer rotates plane-polarized light:

A. To the right (clockwise)

B. It does not rotate light

C. In both directions

D. To the left (counterclockwise)

Q24 Which instrument measures the rotation of plane-polarized light by a chiral compound?

A. Polarimeter

B. Thermometer

C. Barometer

D. Spectrometer

Q25 A compound that rotates plane-polarized light is called optically active.

True

False

Q26 Which property is **different** for a pair of enantiomers?

A. Molecular formula

B. Molecular weight

C. Interaction with plane-polarized light

D. Boiling point

Q27 Which **chemical** property is different for a pair of enantiomers?

A. Their reactions with achiral reagents such as water

B. Their molecular formula

C. Their interactions with other chiral substances or molecules

D. The order of bonding of their atoms

Q28 Which of the following is **NOT** a property of enantiomers?

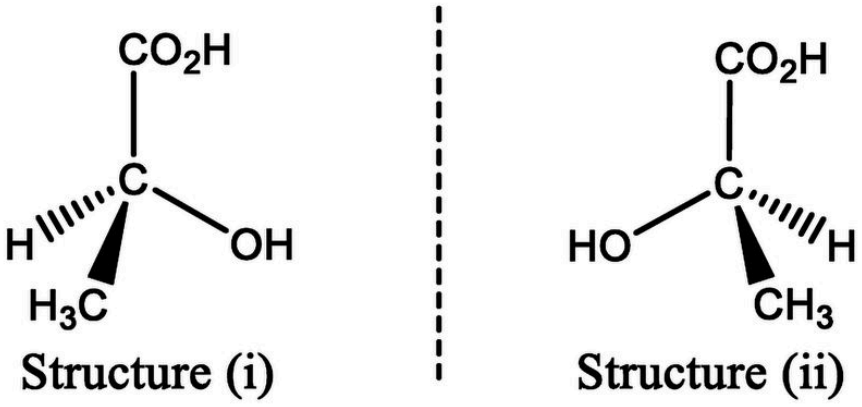
A. They differ in their interactions with other chiral substances

B. They rotate plane-polarized light in opposite directions

C. They have a plane of symmetry

D. They are non-superimposable mirror images

Q29 L(+)-Lactic acid is produced in muscles during exercise. Where is D(−)-lactic acid found? (The two mirror-image forms of lactic acid are shown below.)



A. Muscles

B. Blood

C. Sour milk

D. Sea water

Q30 Compounds having the same order of bonded atoms but differing only in the direction of rotation of plane-polarized light are:

A. Constitutional isomers

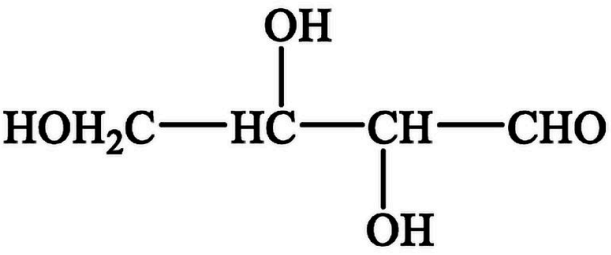
B. Diastereomers

C. Enantiomers

D. Geometric isomers

1.5 Diastereomers & Meso Compounds

Q31 How many stereoisomers are expected for compound (i), 2,3,4-trihydroxybutanal?



(i)

2,3,4-Trihydroxybutanal

A. 3

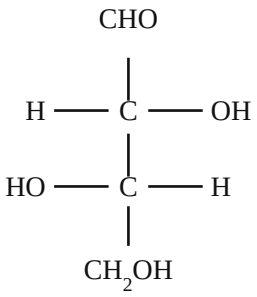
B. 4

C. 2

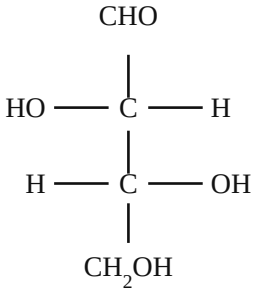
D. 8

Q32 The four stereoisomers of compound (i), 2,3,4-trihydroxybutanal, are shown below as Fischer projections. A and C are:

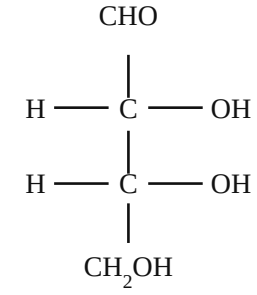
A



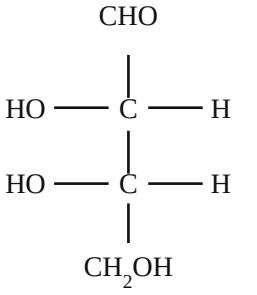
B



C



D



A. The same compound

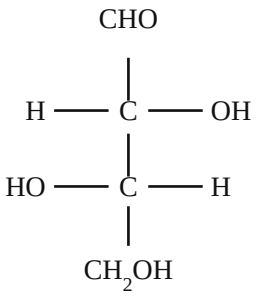
B. Meso compounds

C. Enantiomers

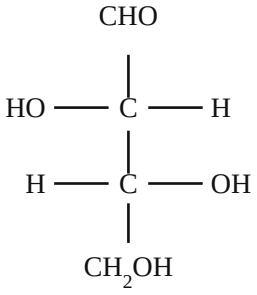
D. Diastereomers

Q33 The four stereoisomers of compound (i), 2,3,4-trihydroxybutanal, are shown below as Fischer projections. A and B are:

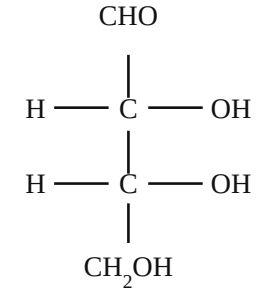
A



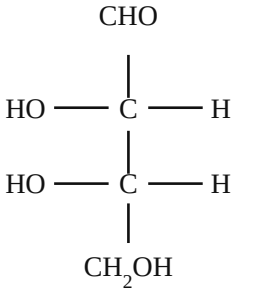
B



C



D



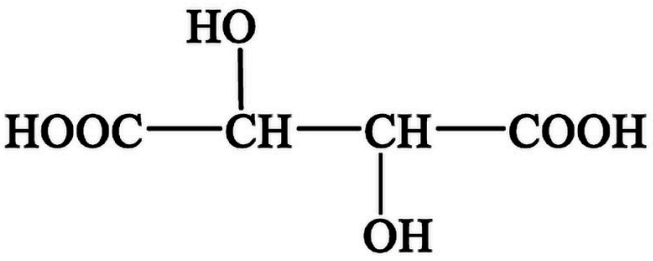
A. Meso compounds

B. The same compound

C. Diastereomers

D. Enantiomers

Q34 How many **different** stereoisomers does compound (ii), 2,3-dihydroxysuccinic acid, have?

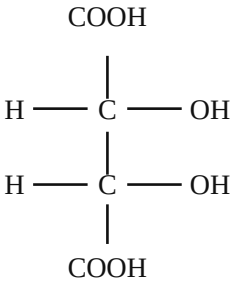


(ii)
2,3-Dihydroxysuccinic acid

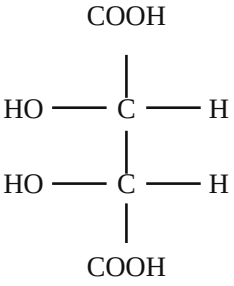
A. 8
B. 3
C. 4
D. 2

Q35 Two of the stereoisomers of compound (ii), 2,3-dihydroxysuccinic acid, are shown below as Fischer projections. G and H are:

G



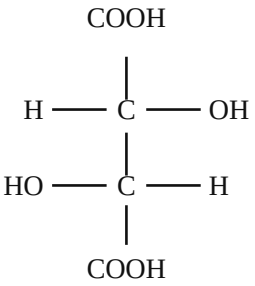
H



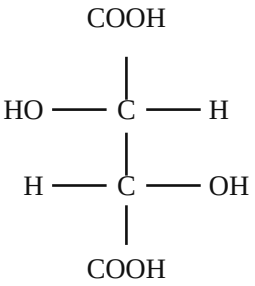
A. Enantiomers
B. Geometric isomers
C. Diastereomers
D. The same meso compound

Q36 Stereoisomers E and F of compound (ii), 2,3-dihydroxysuccinic acid, are shown below. E and F are:

E



F



A. The same meso compound

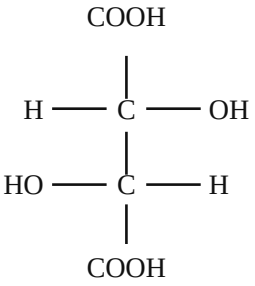
B. Structural isomers

C. Diastereomers

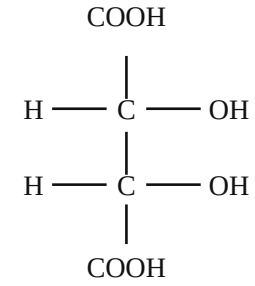
D. Enantiomers

Q37 For compound (ii), 2,3-dihydroxysuccinic acid, what is the relationship between stereoisomer E and the meso form?

E



Meso



A. Enantiomers

B. Diastereomers

C. The same compound

D. Structural isomers

Q38 A meso compound:

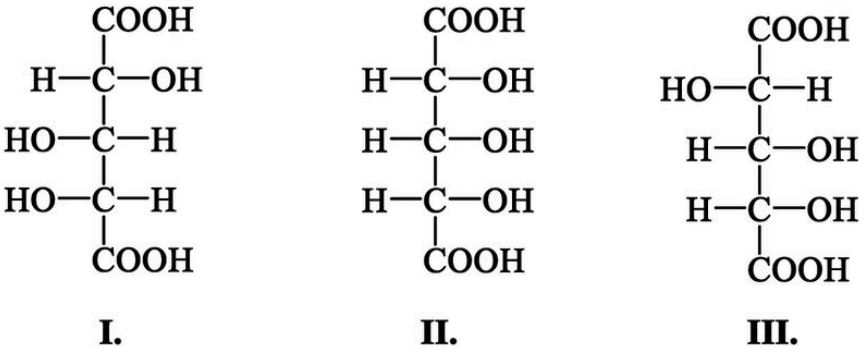
A. Is a pair of enantiomers

B. Has only one stereocenter

C. Rotates plane-polarized light

D. Has an internal plane of symmetry and is optically inactive

Q39 Which Fischer projection is the **meso** form?



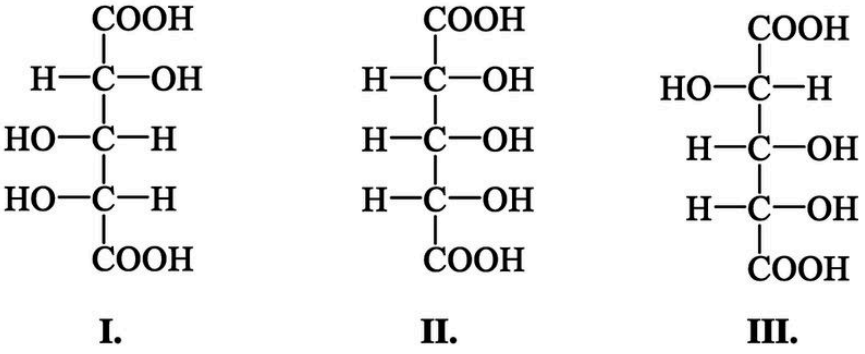
A. None

B. II

C. III

D. I

Q40 In the Fischer projections shown below, which pair are enantiomers?



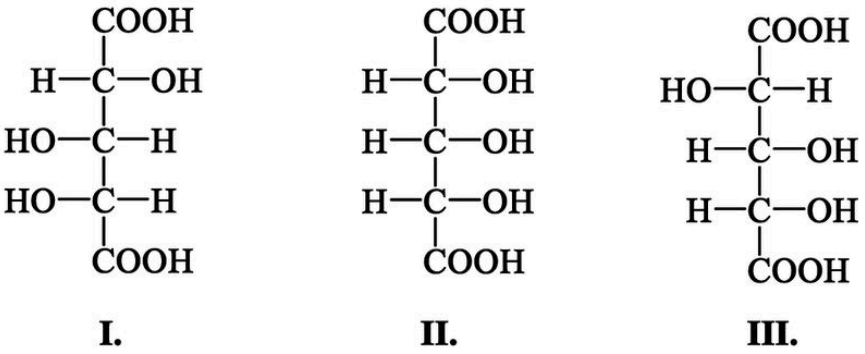
A. I and II

B. II and III

C. I and III

D. None

Q41 In the Fischer projections shown below, which pairs are **diastereomers**?



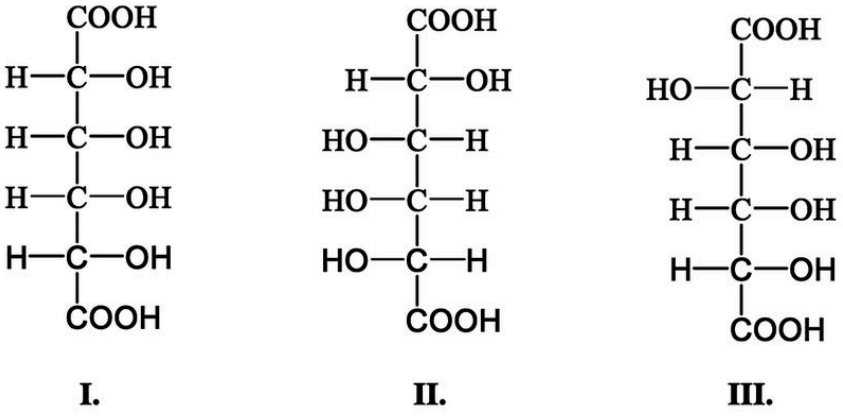
A. II and III only

B. I and II, and I and III

C. I and II, and II and III

D. I and III only

Q42 Which molecule is the **meso** form?



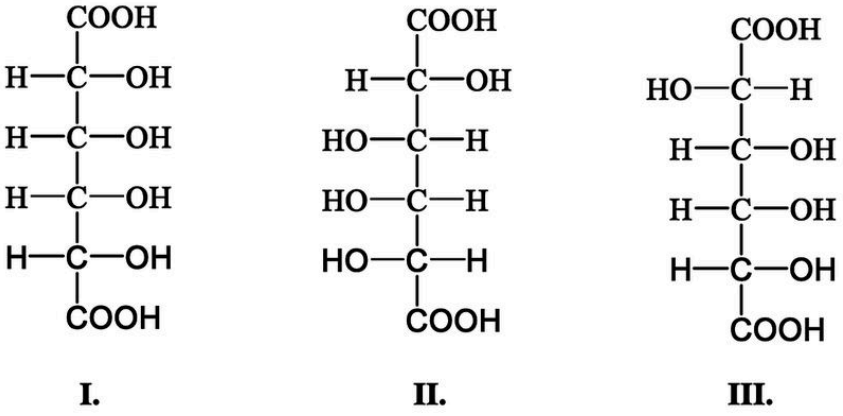
A. I

B. None

C. III

D. II

Q43 In the Fischer projections shown below, II and III are:



A. Enantiomers

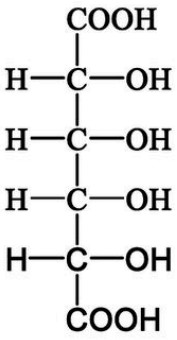
B. Diastereomers

C. Meso forms

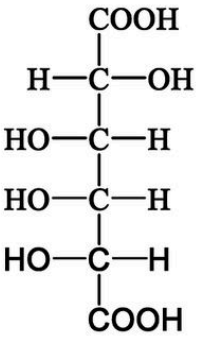
D. Geometric isomers

E. The same compound

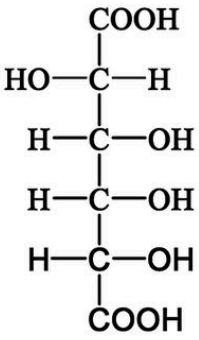
Q44 In the Fischer projections shown below, I and II are:



I.



II.



III.

A. Enantiomers

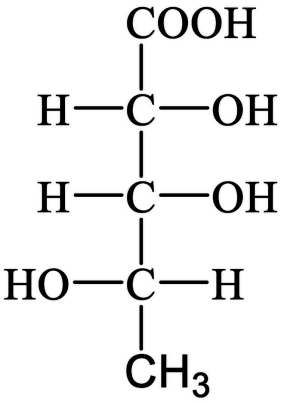
B. Diastereomers

C. Meso forms

D. Geometric isomers

E. The same compound

Q45 How many stereoisomers would you expect for the molecule shown below?



A. 2

B. 4

C. 8

D. 16

E. 32

Answers · Lecture 1

Q1 B	**Q10** D	**Q19** D	**Q28** C	**Q37** B
Q2 A	**Q11** A	**Q20** C	**Q29** C	**Q38** D
Q3 False	**Q12** C	**Q21** B	**Q30** C	**Q39** B
Q4 True	**Q13** A	**Q22** B	**Q31** B	**Q40** C
Q5 A	**Q14** C	**Q23** A	**Q32** D	**Q41** C
Q6 B	**Q15** A	**Q24** A	**Q33** D	**Q42** A
Q7 D	**Q16** B	**Q25** True	**Q34** B	**Q43** A
Q8 D	**Q17** A	**Q26** C	**Q35** D	**Q44** B
Q9 A	**Q18** True	**Q27** C	**Q36** D	**Q45** C

2 Geometric Isomerism

In this lecture

1. Geometric Isomerism (cis / trans) 10 questions
2. E / Z System & Priority Rules 15 questions
3. Geometric Isomerism in Cyclic Compounds 6 questions
4. Conformations of Open-Chain Compounds 11 questions
5. Conformations of Cyclohexane 8 questions

2.1 Geometric Isomerism (cis / trans)

Q1 Geometric isomerism is a result of:

A. Restricted rotation (rigidity) about a double bond or in a ring

B. The presence of a chiral carbon

C. A different order of bonded atoms

D. Different molecular formulas

Q2 Geometric isomerism occurs in:

A. Alkynes only

B. Alkenes and cyclic compounds

C. Alkanes only

D. Compounds with a chiral carbon only

Q3 Geometric isomers have the same molecular formula and the same order of bonded atoms.

True

False

Q4 A molecule is designated *cis* when the two groups are:

A. On the same carbon atom

B. On opposite sides of the double bond

C. On the same side of the double bond

D. Perpendicular to the double bond

Q5 Which statement about the two isomers of 1,2-dichloroethene is correct?

A. They have different molecular formulas

B. trans-1,2-Dichloroethene boils at 60 °C, higher than the cis isomer

C. Both isomers have the same boiling point

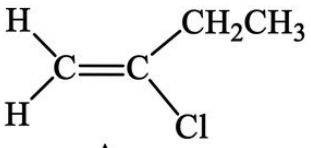
D. cis-1,2-Dichloroethene boils at 60 °C, higher than the trans isomer (48 °C)

Q6 Rotation about a C–C sigma (single) bond is free, but rotation about a C=C double bond is restricted.

True

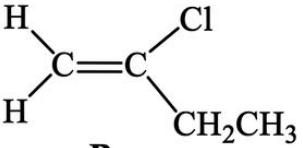
False

Q7 Molecules A and B are:



A.

and



B.

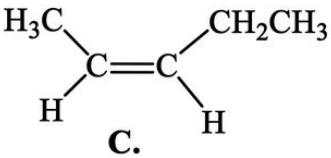
A. The same molecule

B. cis/trans isomers

C. Structural isomers

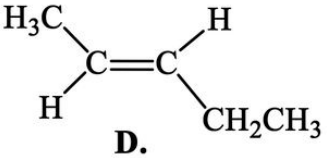
D. Enantiomers

Q8 Molecules C and D are:



C.

and



D.

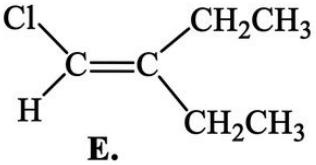
A. Not isomers

B. Geometric isomers: cis-2-pentene and trans-2-pentene

C. The same molecule

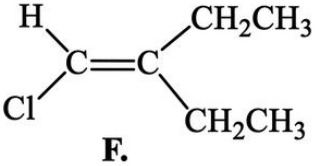
D. Structural isomers

Q9 Molecules E and F are:



E.

and



F.

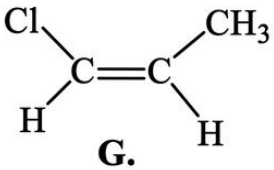
A. cis/trans isomers

B. Enantiomers

C. Structural isomers

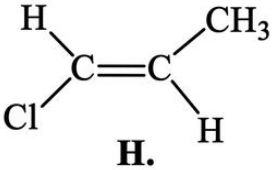
D. The same molecule

Q10 Molecules G and H are:



G.

and



H.

A. The same molecule

B. Structural isomers

C. cis-1-chloropropene and trans-1-chloropropene

D. trans-1-chloropropene and cis-1-chloropropene

2.2 E / Z System & Priority Rules

Q11 The letter **Z** comes from the German word *zusammen*, which means:

A. Left

B. Across

C. Right

D. Together

Q12 The configuration is **(E)** when the higher-priority atoms or groups are:

A. On opposite sides of the C=C double bond

B. On the same carbon atom

C. Both hydrogen atoms

D. On the same side of the C=C double bond

Q13 By sequence rule 1, which atom has the **highest** priority?

A. F

B. I

C. Br

D. Cl

Q14 By sequence rule 2 (for a group of atoms, look at the atomic number of the atom directly attached to the sp² carbon), which group has the **highest** priority?

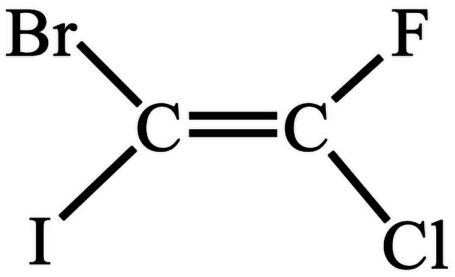
A. –SH

B. –OH

C. $–NH_2$

D. $–CH_3$

Q15 What is the configuration of molecule (ii), 1-bromo-2-chloro-2-fluoro-1-iodoethene?



Molecule (ii)

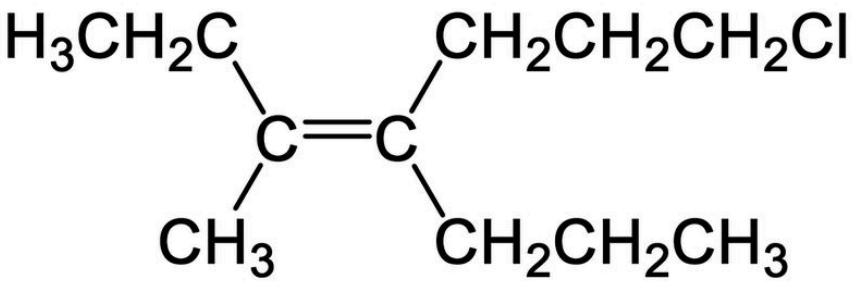
A. (E)

B. It has no E/Z configuration

C. Cannot be decided

D. (Z)

Q16 In molecule (i), which group on the right-hand carbon has the higher priority?



molecule (i)

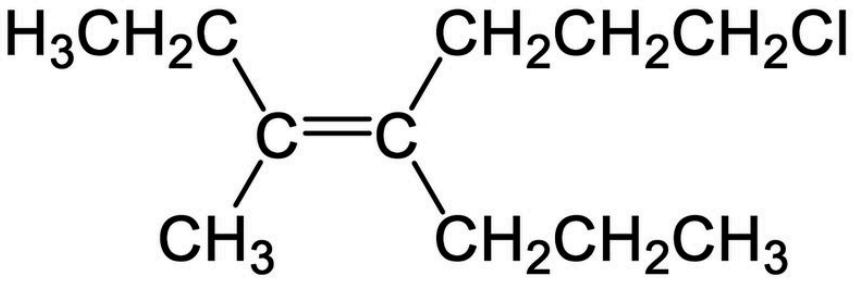
A. It cannot be determined

B. They have equal priority

C. $–CH_2CH_2CH_2Cl$

D. $–CH_2CH_2CH_3$

Q17 What is the configuration of molecule (i), 7-chloro-3-methyl-4-propylhept-3-ene?



molecule (i)

A. cis only, not E/Z

B. It has no E/Z configuration

C. (Z)-7-chloro-3-methyl-4-propylhept-3-ene

D. (E)-7-chloro-3-methyl-4-propylhept-3-ene

Q18 By sequence rule 3 (isotopes), which has the higher priority?

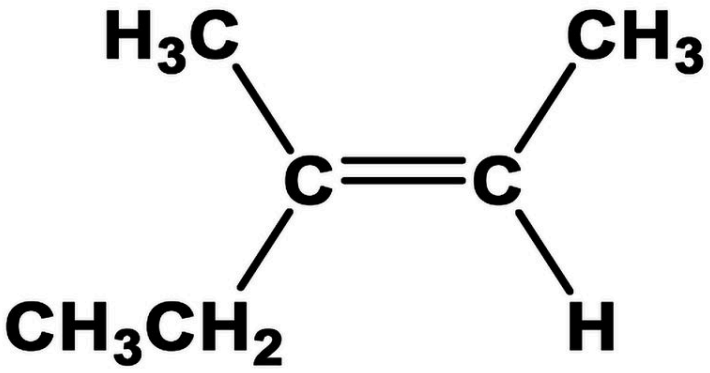
A. Hydrogen (H, ^1H)

B. It depends on the position

C. They are equal

D. Deuterium (D, ^2H)

Q19 In (E)-3-methyl-2-pentene, why does the ethyl group have a higher priority than the methyl group?



A. Ethyl contains a double bond

B. Its carbon carries (C,H,H) while the methyl carbon carries (H,H,H)

C. Methyl has a higher atomic number

D. Ethyl is attached by an oxygen atom

Q20 By sequence rule 5 (multiple bonds), which group has the **highest** priority?

A. –COOH

B. –CN

C. $–CH_2OH$

D. –CHO

Q21 Which list shows the groups in order of **increasing** priority (lowest to highest)?

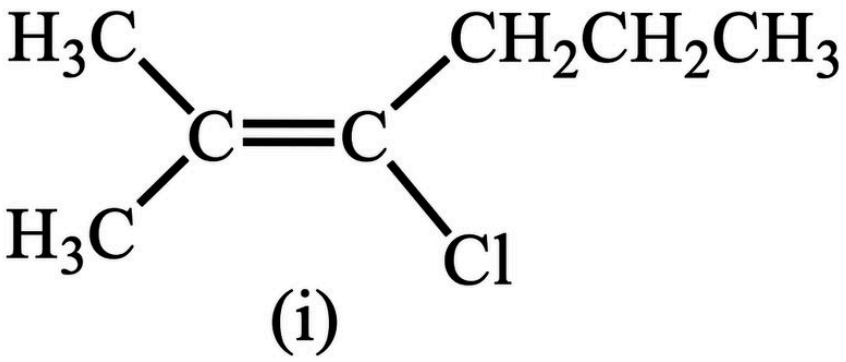
A. $–H < –OH < –CH_2OH < –CO_2H < –CO_2CH_3 < –SCH_3$

B. $–H < –CH_2OH < –CO_2CH_3 < –CO_2H < –OH < –SCH_3$

C. $–H < –CH_2OH < –CO_2H < –CO_2CH_3 < –OH < –SCH_3$

D. $–SCH_3 < –OH < –CO_2CH_3 < –CO_2H < –CH_2OH < –H$

Q22 What is the configuration of compound (i)?



(i)

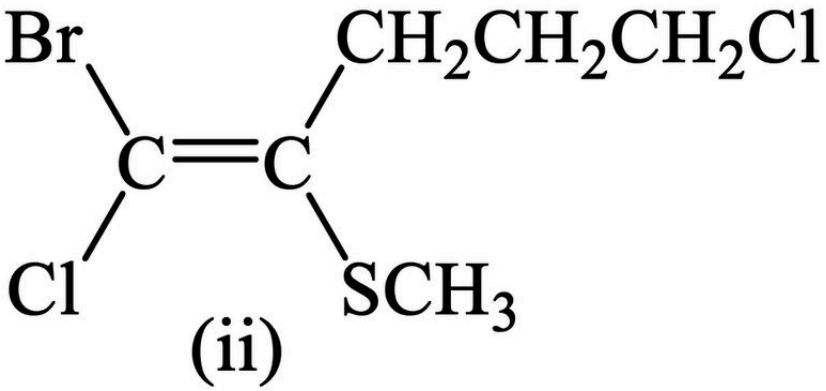
A. (Z)

B. No E/Z configuration

C. cis

D. (E)

Q23 What is the configuration of compound (ii)?



(ii)

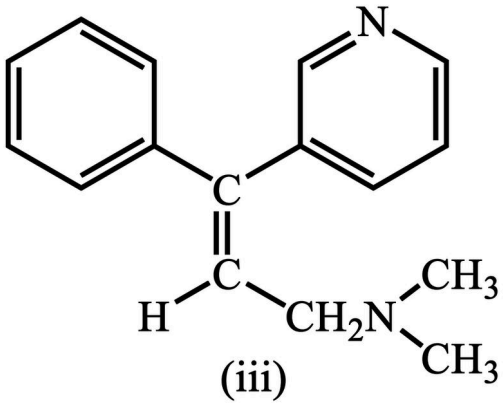
A. cis

B. No E/Z configuration

C. (Z)

D. (E)

Q24 What is the configuration of compound (iii)?



(iii)

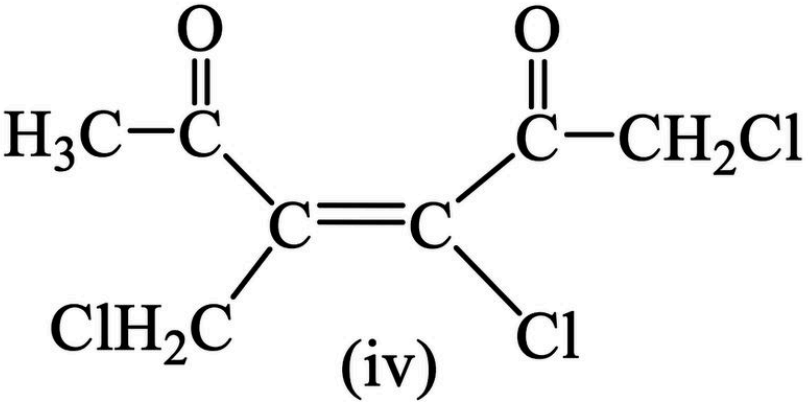
A. (Z)

B. trans

C. (E)

D. No E/Z configuration

Q25 What is the configuration of compound (iv)?



(iv)

A. No E/Z configuration

B. cis only

C. (E)

D. (Z)

2.3 Geometric Isomerism in Cyclic Compounds

Q26 In a wedge-dash drawing of a ring, a solid wedge bond means the group is:

A. Above the plane of the ring

B. Below the plane of the ring

C. In the plane of the ring

D. Always axial

Q27 A dashed bond represents a group below the plane of the ring.

True

False

Q28 In cis-1,2-dimethylcyclopentane, the two CH_3 groups are:

A. In the plane of the ring

B. On the same carbon

C. On opposite sides of the ring

D. On the same side of the ring (both wedges or both dashes)

Q29 In trans-1,4-dimethylcyclohexane drawn with wedge-dash bonds:

A. Both CH_3 groups are wedges

B. One CH_3 is a wedge and the other is a dash

C. Both CH_3 groups are dashes

D. Both CH_3 groups are on the same carbon

Q30 In cis-1,4-dimethylcyclohexane drawn with wedge-dash bonds:

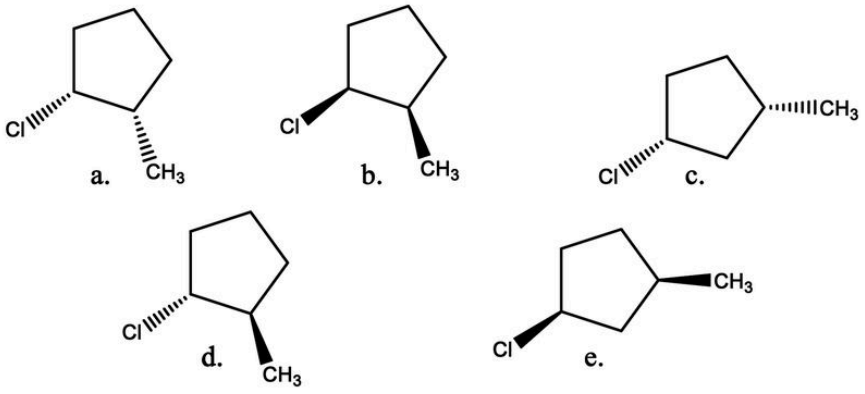
A. One CH_3 is drawn in the plane of the ring and the other above it

B. One CH_3 is a wedge and the other is a dash

C. Both CH_3 groups are on the same side of the ring (both wedges, or both dashes)

D. Both CH_3 groups are on the same carbon

Q31 The structure of *trans*-1-chloro-2-methylcyclopentane is:



A. a

B. b

C. c

D. d

E. e

2.4 Conformations of Open-Chain Compounds

Q32 Which of the following is **NOT** a projection formula used to show conformations?

A. Lewis electron-dot formula

B. Wedge-dash formula

C. Newman projection

D. Sawhorse projection

Q33 Which projection looks straight down the C–C bond, with the front carbon as a point and the back carbon as a circle?

A. Fischer projection

B. Ball-and-stick formula

C. Newman projection

D. Sawhorse projection

Q34 Which of the projection / dimensional formulas in the doctor's answer draws the molecule with crossing horizontal and vertical lines, the carbon atoms being at the crossing points?

A. Fischer projection formula

B. Wedge-dash formula

C. Sawhorse projection formula

D. Ball-and-stick formula

Q35 Which conformer of butane is the most stable?

A. Conformer with eclipsed methyl groups

B. Anti (staggered) conformer

C. Gauche conformer

D. Eclipsed conformer (CH_3 eclipsing H)

Q36 Which conformer of butane is the **least** stable?

A. Eclipsed (CH_3 eclipsing H)

B. Gauche

C. Anti (staggered)

D. Eclipsed methyls (CH_3 eclipsing CH_3)

Q37 What is the order of stability of the butane conformers?

A. Eclipsed methyls > eclipsed > gauche > anti

B. Anti > eclipsed > gauche > eclipsed methyls

C. Anti > gauche > eclipsed > eclipsed methyls

D. Gauche > anti > eclipsed > eclipsed methyls

Q38 In the gauche conformation of butane, the two CH_3 groups are:

A. 60° apart (staggered)

B. 0° apart (eclipsed)

C. 180° apart

D. 120° apart (eclipsed)

Q39 What is the major difference between configuration and conformation?

A. Conformation is fixed in space; configuration changes by rotation

B. Configuration depends only on the molecular formula

C. Configuration is fixed in space; conformation changes by rotation about single bonds

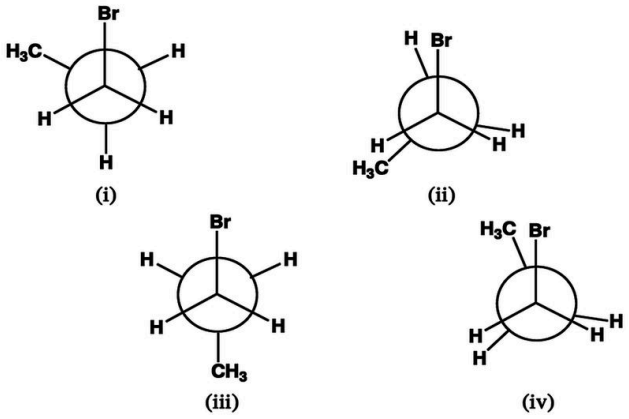
D. They are the same thing

Q40 Conformers can be interconverted by rotation about single bonds without breaking any bonds.

True

False

Q41 Arrange the following Newman projections in order of **decreasing** energy (highest to lowest):



A. (i) > (ii) > (iii) > (iv)

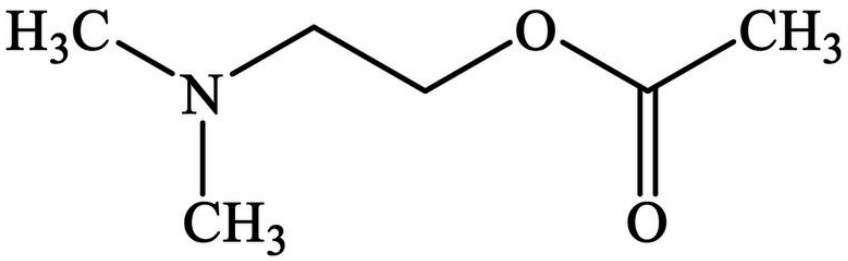
B. (iii) > (i) > (ii) > (iv)

C. (ii) > (iii) > (iv) > (i)

D. (iii) > (iv) > (ii) > (i)

E. (iv) > (ii) > (i) > (iii)

Q42 Draw all the possible conformers of the compound shown below (looking along the central CH_2–CH_2 bond). Which conformer is the **most stable**?



A. All conformers have the same energy

B. The anti (staggered) conformer, with –$N(CH_3)_2$ and –$OC(O)CH_3$ 180° apart

C. The eclipsed conformer, with –$N(CH_3)_2$ and –$OC(O)CH_3$ eclipsing each other

D. The gauche (staggered) conformer, with –$N(CH_3)_2$ and –$OC(O)CH_3$ 60° apart

2.5 Conformations of Cyclohexane

Q43 Which is the most stable conformation of cyclohexane?

A. Planar hexagon

B. Chair

C. Boat

D. Fully eclipsed

Q44 The boat conformation of cyclohexane is more stable than the chair conformation.

True

False

Q45 In the chair conformation, axial bonds are:

A. Always pointing down

B. Always pointing up

C. Parallel to the ring axis, pointing straight up or straight down

D. Roughly in the plane of the ring, pointing outward

Q46 In chair cyclohexane, each carbon carries:

A. Only one hydrogen

B. Two equatorial hydrogens

C. Two axial hydrogens

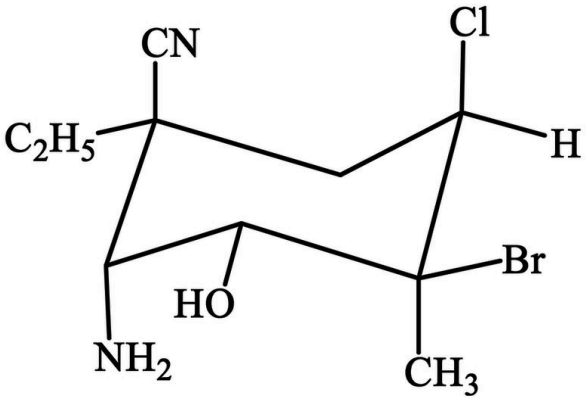
D. One axial and one equatorial hydrogen

Q47 Axial positions alternate up and down around the cyclohexane chair.

True

False

Q48 Which two atoms or groups are in the axial position and above the plane of the molecule?



A. Cl and CH_3

B. CN and NH_2

C. Br and Cl

D. NH_2 and OH

E. CN and Cl

Q49 In the chair conformations of trans-1,2-dimethylcyclohexane, the two CH_3 groups are:

A. Always both equatorial

B. Always one axial and one equatorial

C. Both axial (a,a) in one conformer and both equatorial (e,e) in the other

D. Always both axial

Q50 Which chair conformer of trans-1,2-dimethylcyclohexane is more stable?

A. The boat conformer

B. e,e (both CH_3 equatorial)

C. Both are equally stable

D. a,a (both CH_3 axial)

Answers · Lecture 2

Q1 A	**Q11** D	**Q21** C	**Q31** D	**Q41** E
Q2 B	**Q12** A	**Q22** B	**Q32** A	**Q42** B
Q3 True	**Q13** B	**Q23** D	**Q33** C	**Q43** B
Q4 C	**Q14** A	**Q24** A	**Q34** A	**Q44** False
Q5 D	**Q15** D	**Q25** D	**Q35** B	**Q45** C
Q6 True	**Q16** C	**Q26** A	**Q36** D	**Q46** D
Q7 A	**Q17** C	**Q27** True	**Q37** C	**Q47** True
Q8 B	**Q18** D	**Q28** D	**Q38** A	**Q48** E
Q9 D	**Q19** B	**Q29** B	**Q39** C	**Q49** C
Q10 C	**Q20** A	**Q30** C	**Q40** True	**Q50** B

3 Assignment of Configuration

In this lecture

1. Absolute Configuration & Priority Rules 9 questions
2. Drawing Enantiomers: Wedge-Dash & Fischer 9 questions
3. Assigning (R) and (S) 10 questions
4. Stereoisomers, Meso Forms & Relationships 18 questions
5. Racemic Mixtures & Resolution 15 questions
6. Key Terms & Classification of Isomers 14 questions

3.1 Absolute Configuration & Priority Rules

Q1 What does “absolute configuration” around a chiral carbon mean?

A. The order of arrangement in space of the four groups around the chiral carbon

B. The number of chiral carbons in the molecule

C. The direction in which the compound rotates plane-polarized light

D. The molecular formula of the compound

Q2 What is the **first** step in assigning (R) or (S) to a chiral carbon?

A. Measure the optical rotation

B. Draw a curved arrow from group 1 to group 3

C. Determine the molecular formula

D. Identify the chiral carbon and rank its four groups by the Cahn-Ingold-Prelog sequence rules

Q3 After ranking the four groups by the Cahn-Ingold-Prelog rules, what is the next step in assigning (R) or (S)?

A. Count the number of possible stereoisomers using 2^n

B. Project the molecule so that the group with the highest priority is in the rear

C. Project the molecule so that the group with the lowest priority is in the rear

D. Measure the optical rotation of the compound with a polarimeter

Q4 With the lowest-priority group in the rear, how is the curved arrow drawn?

A. From the group of lowest priority to the highest priority

B. From the group of highest priority to the next highest priority, then to the next ($1 \rightarrow 2 \rightarrow 3$)

C. From the lowest-priority group to the chiral carbon

D. From priority $3 \rightarrow 2 \rightarrow 1$

Q5 With the lowest-priority group pointing away from the viewer, the arrow $1 \rightarrow 2 \rightarrow 3$ goes **clockwise**. The configuration is:

A. (R)

B. (S)

Q6 By rule 1, which order of priority is correct?

A. I > Br > Cl > S > F > O > N > C > H

B. H > C > N > O > F > S > Cl > Br > I

C. F > Cl > Br > I > O > N > C > H

D. I > Br > Cl > F > S > O > N > C > H

Q7 By rule 2, which group has the **highest** priority?

A. $–CH_2NH_2$

B. $–CH_2OCH_3$

C. $–CH_2SCH_2CH_3$

D. $–CH_2SH$

Q8 By rule 3, how are the atoms of a double or triple bond treated?

A. They are counted only once

B. As duplicated (double bond) or triplicated (triple bond) atoms

C. They always get the lowest priority

D. They are ignored

Q9 The priority order of the groups on the chiral carbon of 1-bromo-1-chloroethane is:

A. Cl > Br > CH_3 > H

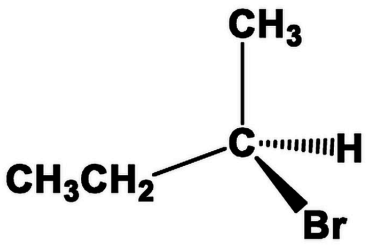
B. CH_3 > Br > Cl > H

C. Br > CH_3 > Cl > H

D. Br > Cl > CH_3 > H

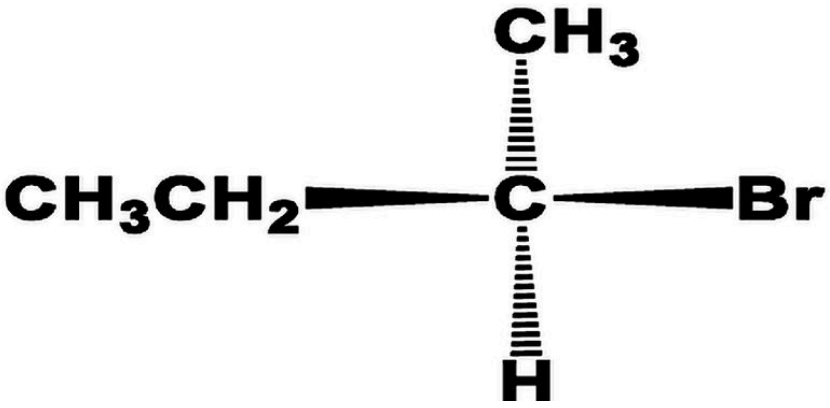
3.2 Drawing Enantiomers: Wedge-Dash & Fischer

Q10 Which structure is the mirror image of 2-bromobutane (a)?

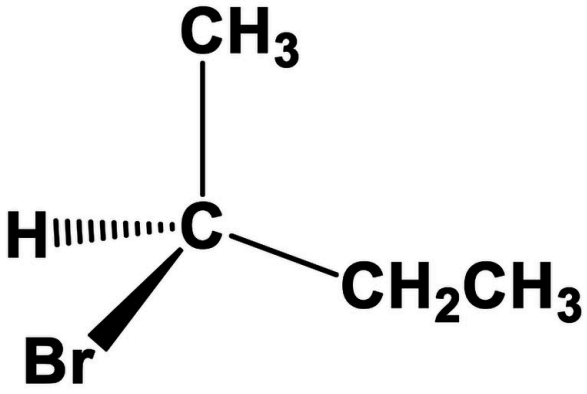


(a)
Tetrahedral structure
of 2-Bromobutane

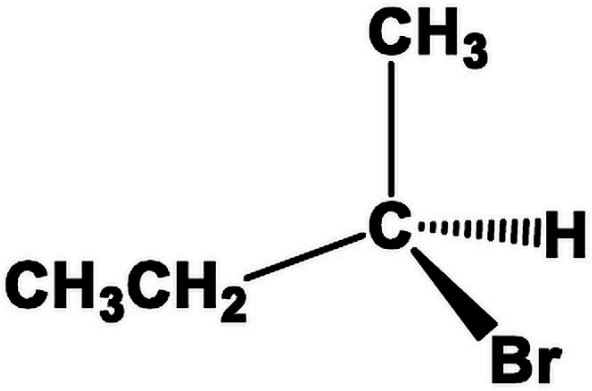
A.



B.

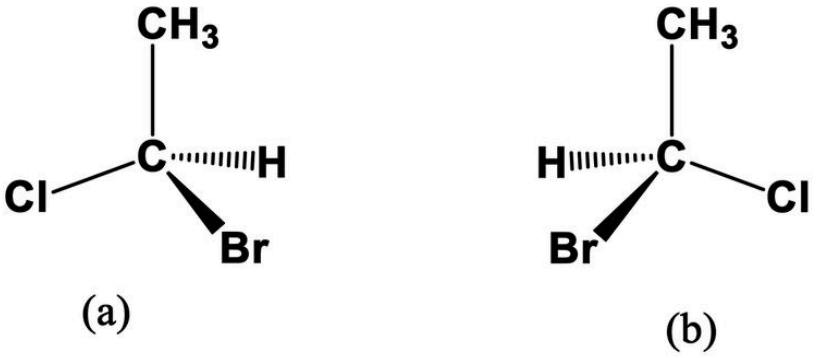


C.



D. None of these

Q11 Structures (a) and (b) of 1-bromo-1-chloroethane shown below are:



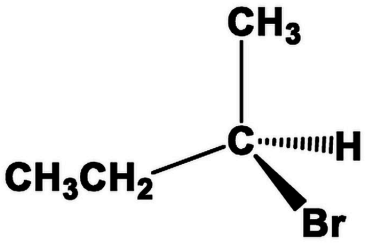
A. The same molecule

B. Structural isomers

C. Diastereomers

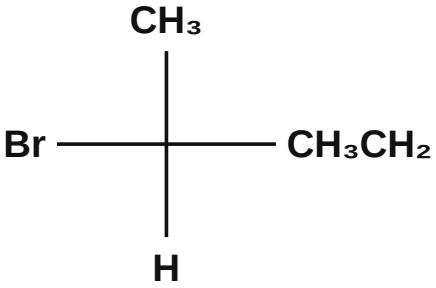
D. Enantiomers

Q12 Which Fischer projection represents 2-bromobutane (a)?

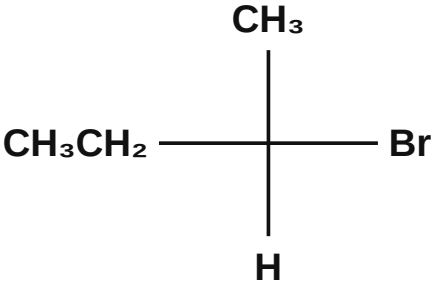


(a)
Tetrahedral structure
of 2-Bromobutane

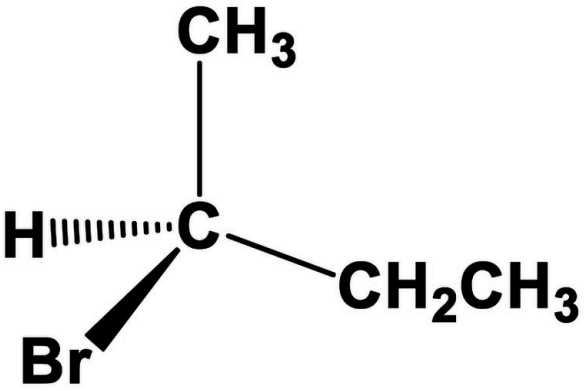
A.



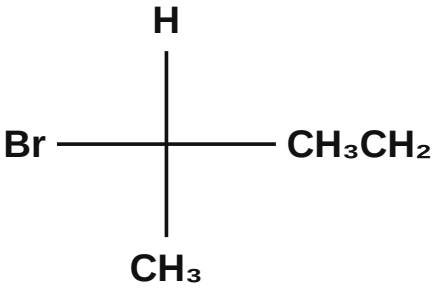
B.



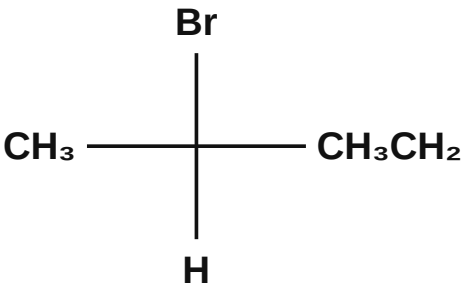
Q13 Which Fischer projection represents 2-bromobutane (b), the mirror image of (a), shown below?



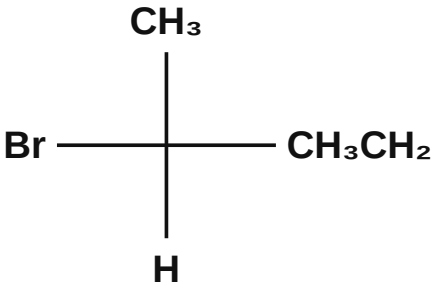
A.



B.



C.



Q14 In a Fischer projection, the vertical bonds point _____ the viewer and the horizontal bonds point _____ the viewer.

A. away from … towards

B. towards … away from

C. away from … away from

D. towards … towards

Q15 A Fischer projection of a chiral carbon has the lowest-priority group on a horizontal bond. To assign (R)/(S) properly, the doctor's method says:

A. Swap it with a vertical group and leave the other two groups unchanged

B. Rotate the whole drawing by 90° so that the group becomes vertical

C. Swap it with a vertical group, then swap the other two groups to keep the same configuration

D. Ignore its position and read the arrow 1 → 2 → 3 directly as drawn

Q16 Interchanging only **one pair** of groups around a chiral carbon gives:

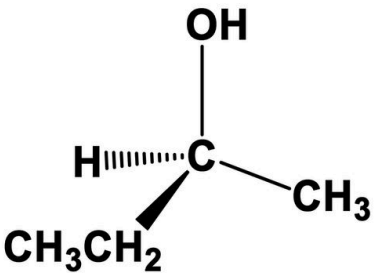
A. A meso compound

B. The enantiomer (mirror image)

C. The same molecule

D. A structural isomer

Q17 What is the configuration of the 2-butanol shown?

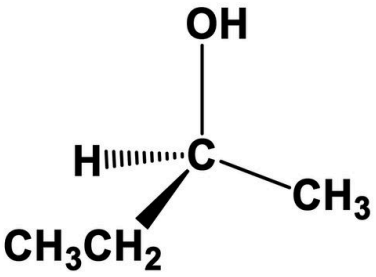


Wedge-Dash (tetrahedral structure)
of Butan-2-ol

A. (S)

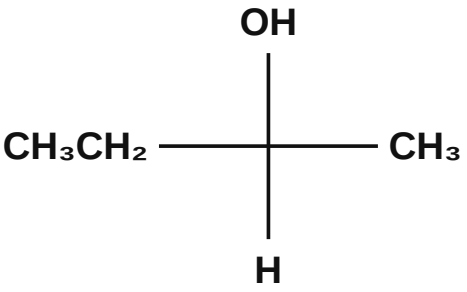
B. (R)

Q18 Which Fischer projection matches the 2-butanol shown?

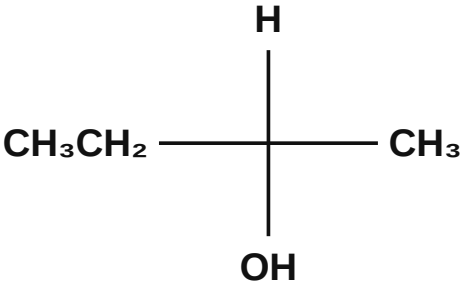


Wedge-Dash (tetrahedral structure) of Butan-2-ol

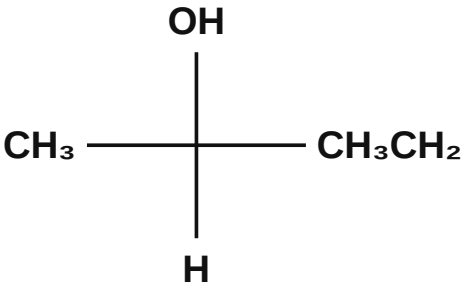
A.



B.

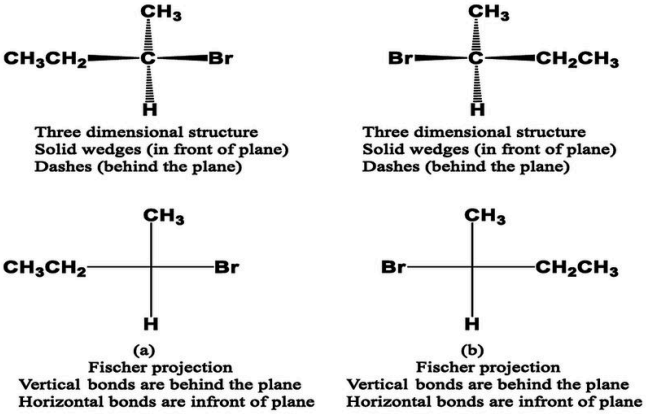


C.



3.3 Assigning (R) and (S)

Q19 What are the configurations of 2-bromobutane (a) and (b)?



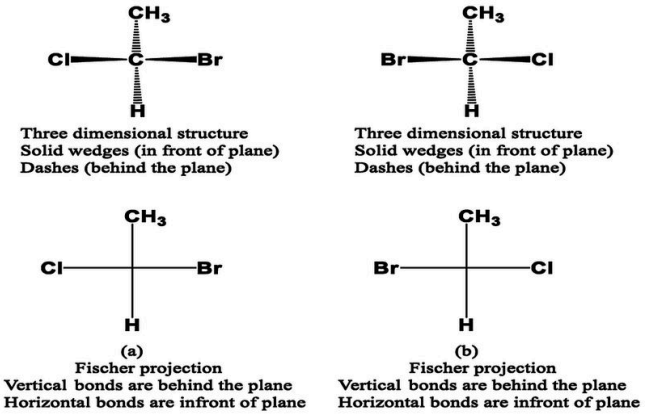
A. (a) is (S) and (b) is (R)

B. (a) is (R) and (b) is (S)

C. Both are (R)

D. Both are (S)

Q20 What are the configurations of 1-bromo-1-chloroethane (a) and (b)?



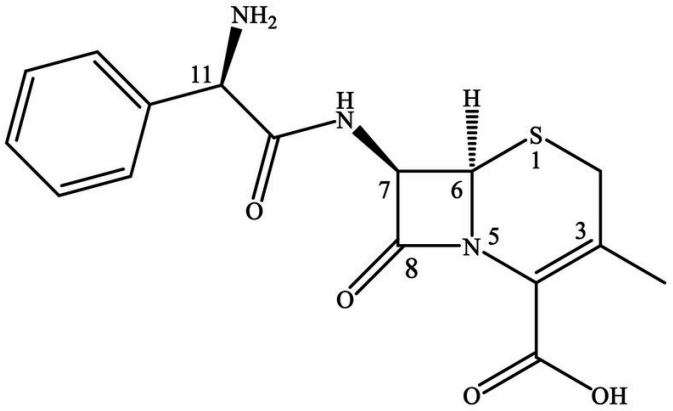
A. Both are (R)

B. Both are (S)

C. (a) is (R) and (b) is (S)

D. (a) is (S) and (b) is (R)

Q21 Assign the configuration at C-6, C-7 and C-11 of the compound shown:



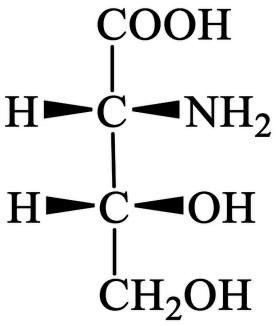
A. (6R, 7R, 11R)

B. (6R, 7S, 11S)

C. (6S, 7R, 11S)

D. (6S, 7S, 11R)

Q22 What is the configuration of molecule (I)?



(I)

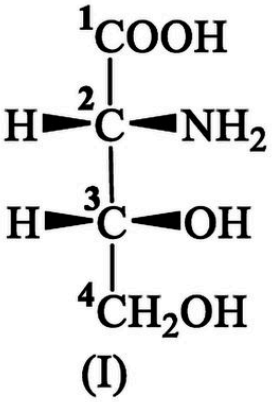
A. (2S, 3R)

B. (2S, 3S)

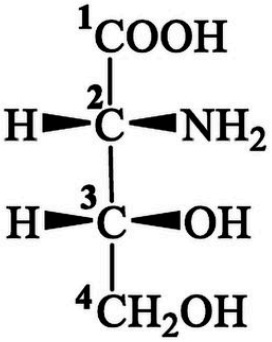
C. (2R, 3R)

D. (2R, 3S)

Q23 Assign the configurations of I and II and comment on their relationship:



(I)



(II)

A. (2R, 3S), I and II are diastereomers

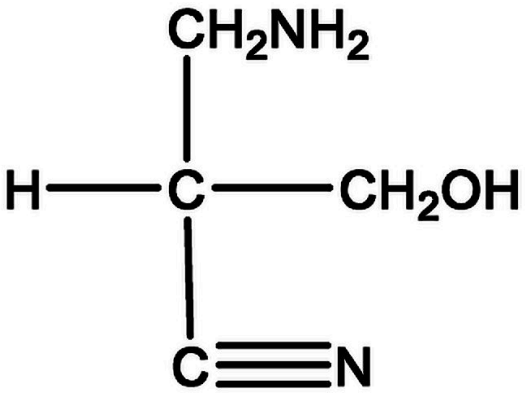
B. (2S, 3S), I and II are meso forms

C. (2S, 3S), I and II are the same molecule

D. (2R, 3S), I and II are enantiomers

E. (2R, 3S), I and II are the same molecule

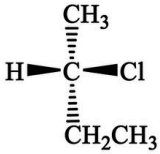
Q24 The absolute configuration of the molecule shown is:



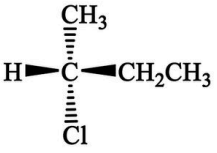
A. (R)

B. (S)

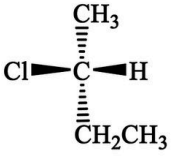
Q25 Which two compounds are the **same** compound?



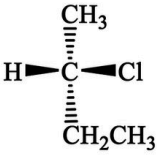
(a)



(b)



(c)



(d)

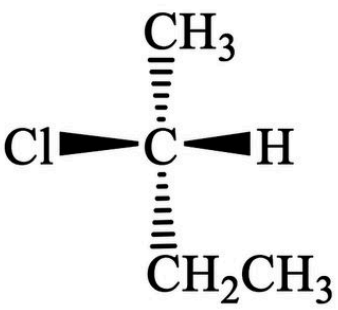
A. (a) and (d)

B. (c) and (d)

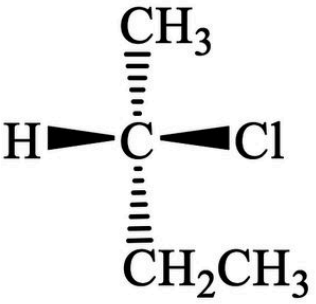
C. (a) and (b)

D. (a) and (c)

Q26 Compounds (c) and (d) shown below are:



(c)



(d)

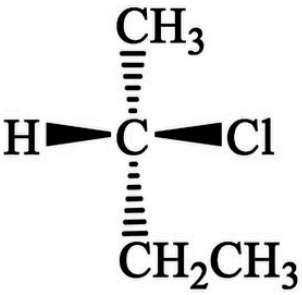
A. Diastereomers

B. The same compound

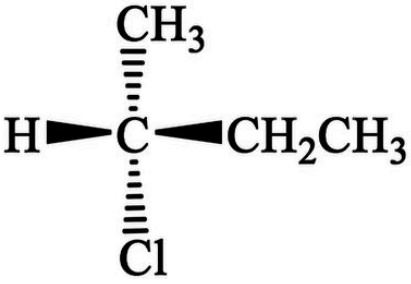
C. Structural isomers

D. Enantiomers

Q27 Compounds (a) and (b) shown below are:



(a)



(b)

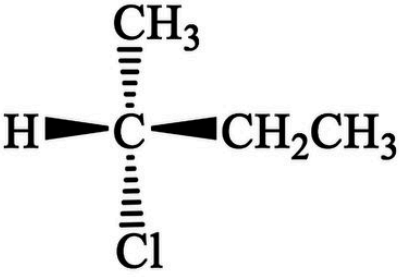
A. Diastereomers

B. Enantiomers

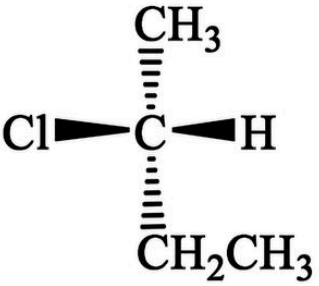
C. The same compound

D. Structural isomers

Q28 Compounds (b) and (c) shown below are:



(b)



(c)

A. Enantiomers

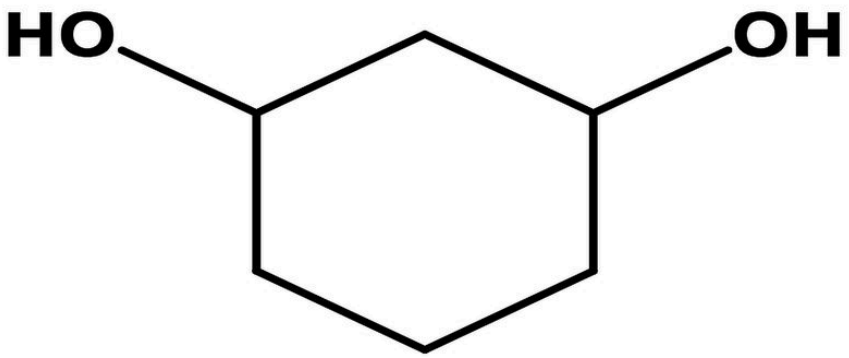
B. Structural isomers

C. The same compound

D. Diastereomers

3.4 Stereoisomers, Meso Forms & Relationships

Q29 How many **different** stereoisomers exist for cyclohexane-1,3-diol?



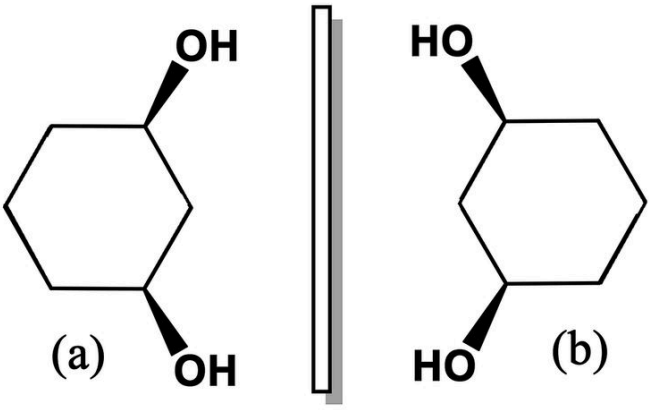
A. 3

B. 8

C. 4

D. 2

Q30 cis-Cyclohexane-1,3-diol, shown below as (a) and its mirror image (b), is:



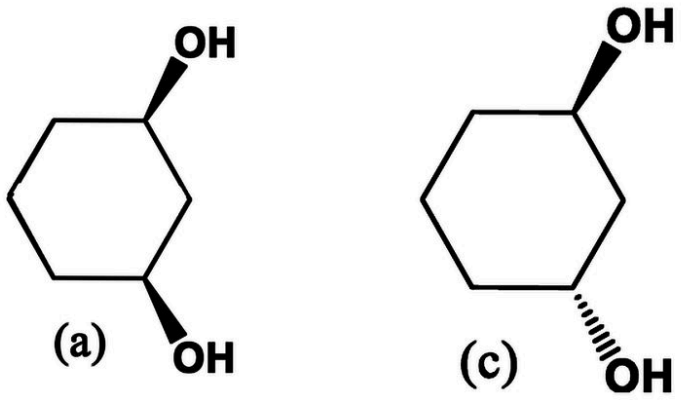
A. Optically active

B. A pair of enantiomers

C. A structural isomer of the trans form

D. A meso compound

Q31 In cyclohexane-1,3-diol, the trans isomer (c) and the meso form are:



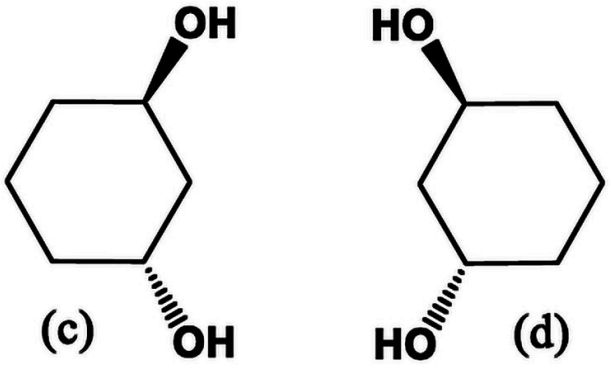
A. Enantiomers

B. Diastereomers

C. Structural isomers

D. The same compound

Q32 In cyclohexane-1,3-diol, the trans isomers (c) and (d) shown below are:



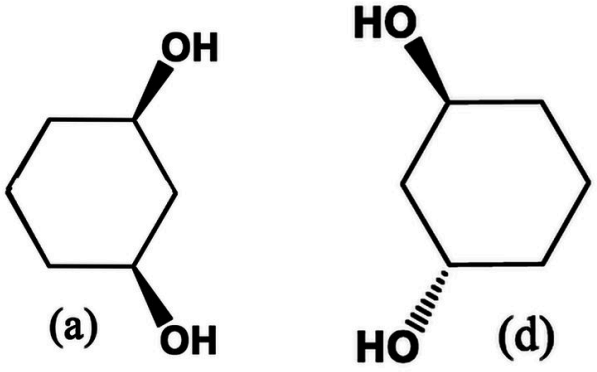
A. The same meso compound

B. Structural isomers

C. Enantiomers

D. Diastereomers

Q33 In cyclohexane-1,3-diol, the meso form (a) and the trans isomer (d) shown below are:



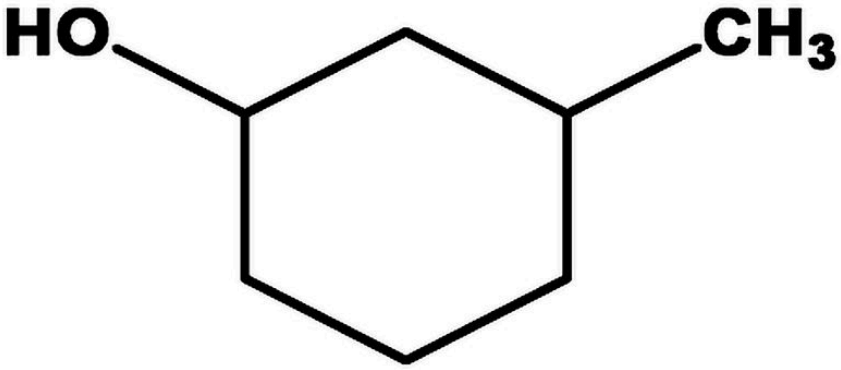
A. Diastereomers

B. The same compound

C. Enantiomers

D. Structural isomers

Q34 How many stereoisomers exist for 3-methylcyclohexan-1-ol?



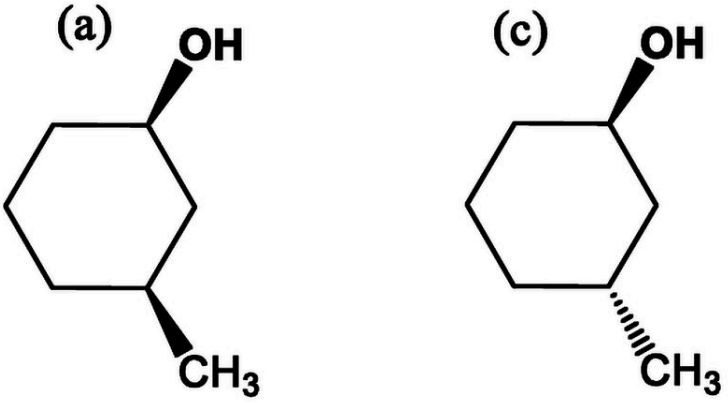
A. 8

B. 3

C. 2

D. 4

Q35 In 3-methylcyclohexan-1-ol, isomers (a) and (c) are:



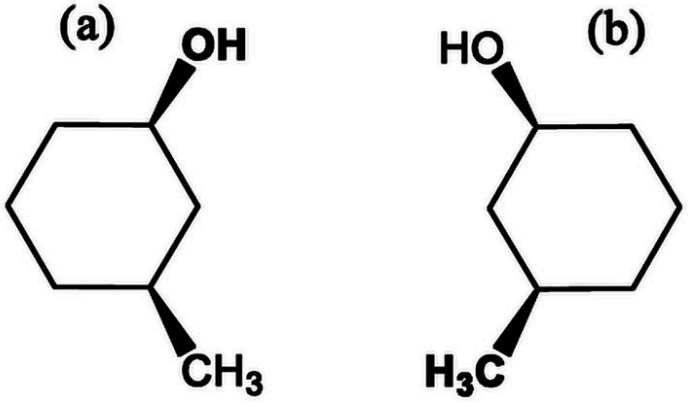
A. The same compound

B. Diastereomers

C. Meso forms

D. Enantiomers

Q36 In 3-methylcyclohexan-1-ol, isomers (a) and (b) are:



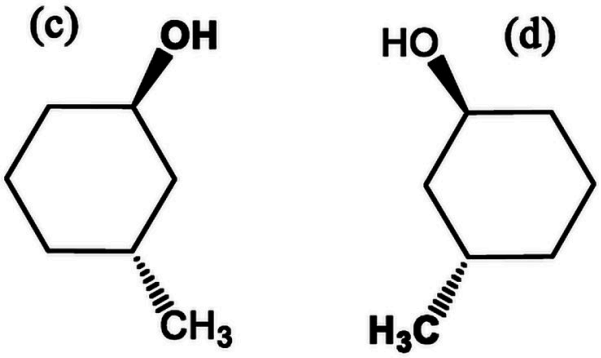
A. Meso forms

B. Enantiomers

C. Diastereomers

D. The same compound

Q37 In 3-methylcyclohexan-1-ol, isomers (c) and (d) are:



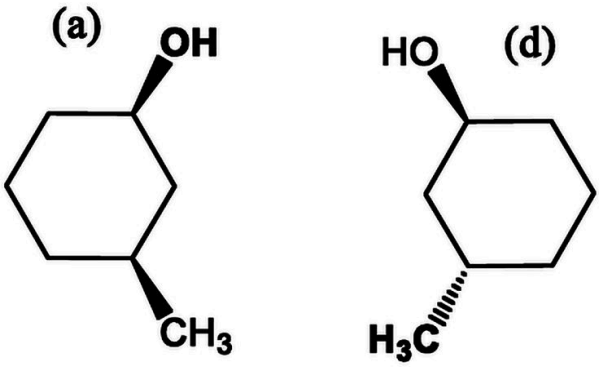
A. The same compound

B. Diastereomers

C. Enantiomers

D. Meso forms

Q38 In 3-methylcyclohexan-1-ol, isomers (a) and (d) are:



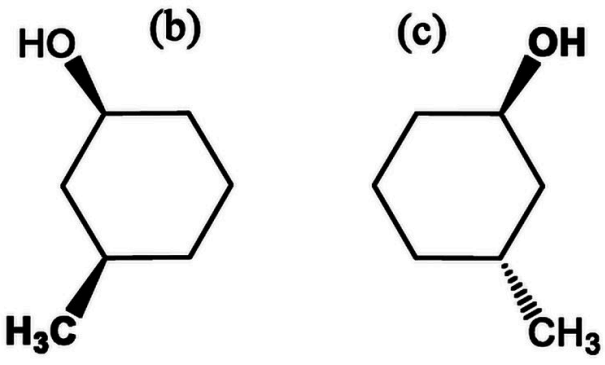
A. Meso forms

B. Enantiomers

C. The same compound

D. Diastereomers

Q39 In 3-methylcyclohexan-1-ol, isomers (b) and (c) are:



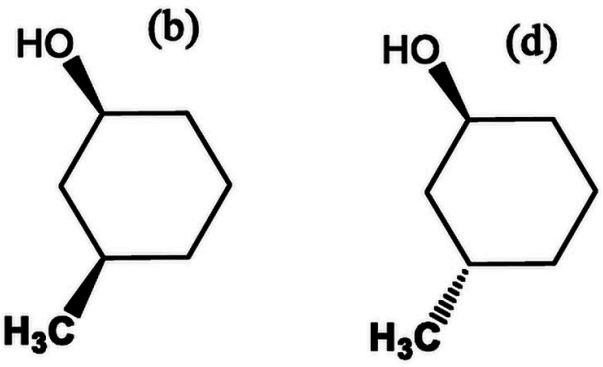
A. Diastereomers

B. Meso forms

C. Enantiomers

D. The same compound

Q40 In 3-methylcyclohexan-1-ol, isomers (b) and (d) are:



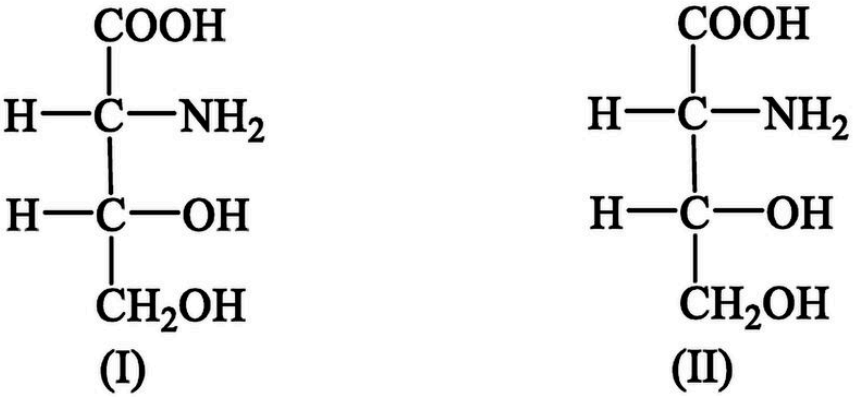
A. Diastereomers

B. The same compound

C. Meso forms

D. Enantiomers

Q41 Molecules (I) and (II) are:



A. The same

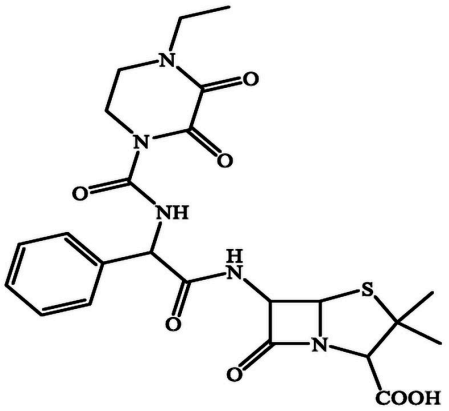
B. Diastereomers

C. Enantiomers

D. Constitutional isomers

E. Optically inactive isomers

Q42 How many chiral carbons are present in the compound shown?



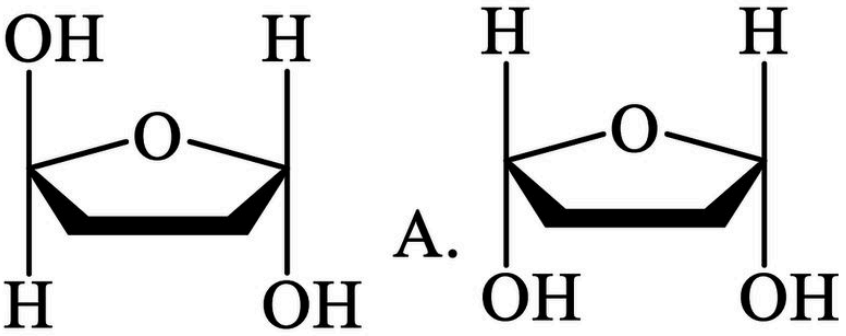
A. 3

B. 4

C. 5

D. 2

Q43 Pair A is best classified as:



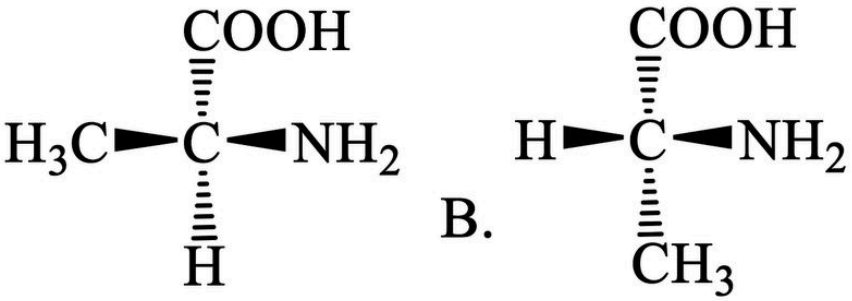
A. Enantiomers

B. Structural isomers

C. Geometric isomers (diastereomers)

D. The same molecule

Q44 Pair B is best classified as:



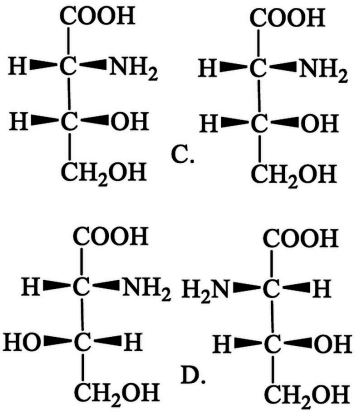
A. The same molecule

B. Diastereomers

C. Geometric isomers

D. Enantiomers

Q45 Pairs C and D are, respectively:



A. Enantiomers; the same molecule

B. Diastereomers; enantiomers

C. The same molecule; enantiomers

D. The same molecule; diastereomers

Q46 In the classification of isomers, geometric isomers are a type of:

A. Structural isomers

B. Conformers

C. Enantiomers

D. Diastereomers

3.5 Racemic Mixtures & Resolution

Q47 A racemic mixture (racemate) is:

A. A mixture of two diastereomers

B. A pure single enantiomer

C. A meso compound

D. A mixture of equal parts of a pair of enantiomers

Q48 Why is a racemic mixture optically inactive?

A. Each of its molecules has an internal plane of symmetry, like a meso form

B. Its molecules contain no chiral carbon, so light passes through unchanged

C. The two enantiomers rotate light equally in opposite directions, so the rotations cancel

D. It absorbs the plane-polarized light instead of rotating it

Q49 Resolution of a racemic mixture means:

A. The physical separation of a racemic mixture into its pure enantiomers

B. Converting one enantiomer into the other

C. Mixing two enantiomers in equal amounts

D. Measuring the specific rotation

Q50 Which of the following is **NOT** one of the three main methods of resolving a racemate?

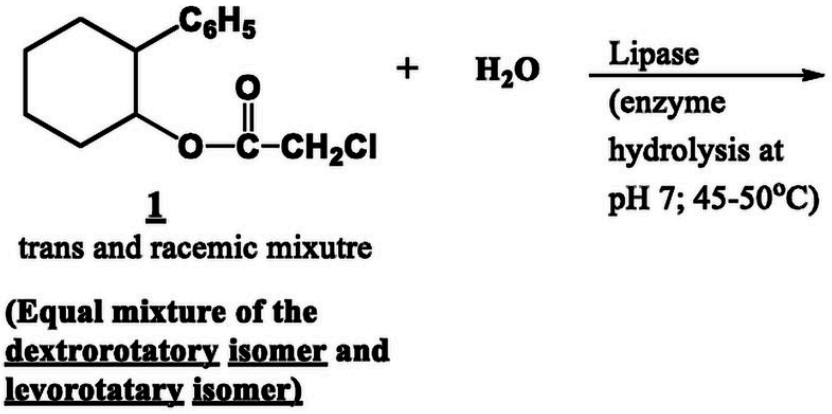
A. Formation of diastereomeric products

B. Simple distillation

C. Chromatography (HPLC) with a chiral stationary phase

D. Enzymatic (biological) resolution

Q51 In the resolution of racemic trans-2-phenylcyclohexyl chloroacetate, the enzyme lipase:



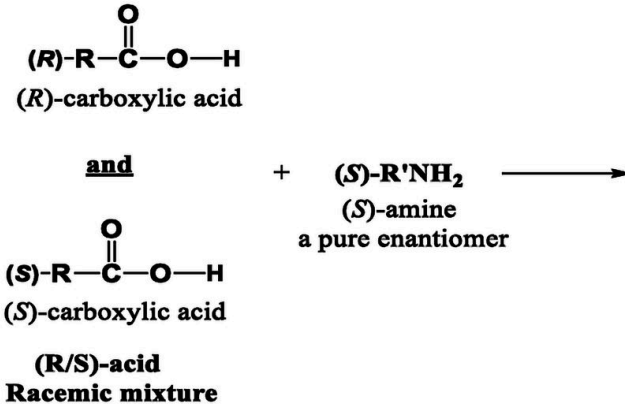
A. Hydrolyzes both enantiomers equally

B. Hydrolyzes the ester of only one enantiomer

C. Does not react with either ester

D. Converts the trans ester into the cis ester

Q52 A racemic (R/S) carboxylic acid is resolved by reacting it with a pure (S)-amine. The products are:



A. A pair of enantiomeric salts that cannot be separated

B. A single meso salt

C. A racemic amine

D. A pair of diastereomeric salts that can be separated by recrystallization

Q53 Which term best describes an enantiomer?

A. Optically active

B. Diastereomer

C. Structural isomer

D. Optically inactive

Q54 A prochiral molecule is:

A. A molecule with chiral carbons and an internal plane of symmetry

B. A molecule that has two or more chiral carbons in its structure

C. A molecule that can undergo a chemical reaction to give a chiral compound

D. A mixture of equal amounts of a pair of enantiomers

Q55 A levorotatory enantiomer rotates plane-polarized light:

A. In both directions

B. It does not rotate light

C. To the right (clockwise), shown as (+)

D. To the left (counterclockwise), shown as (−)

Q56 A dextrorotatory enantiomer rotates plane-polarized light:

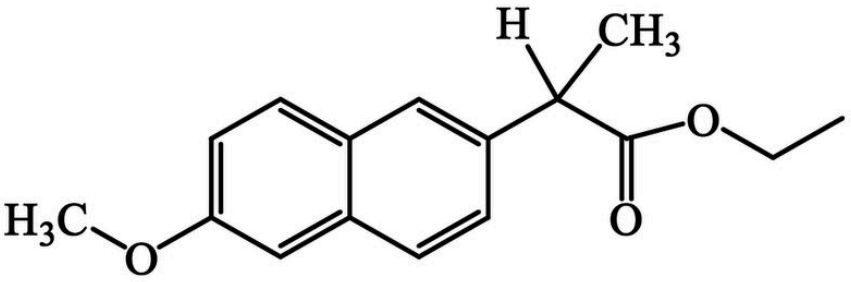
A. To the right (clockwise), shown as (+)

B. It does not rotate light

C. In both directions

D. To the left (counterclockwise), shown as (−)

Q57 Synthesis gave racemic (R/S)-naproxen ethyl ester. Which method gives the active (S)-naproxen?



(R/S)-Naproxen ethyl ester

A. Recrystallization, since the (R)- and (S)-esters have different solubilities

B. Enzymatic hydrolysis with an esterase, which hydrolyzes only the (S)-ester

C. Heating the racemic ester, which converts the (R)-ester into the (S)-ester

D. Simple distillation, since the (R)- and (S)-esters boil at different temperatures

Q58 In the naproxen resolution, how is pure (S)-naproxen (the acid) obtained from its sodium salt?

A. By heating with NaOH

B. By adding more esterase

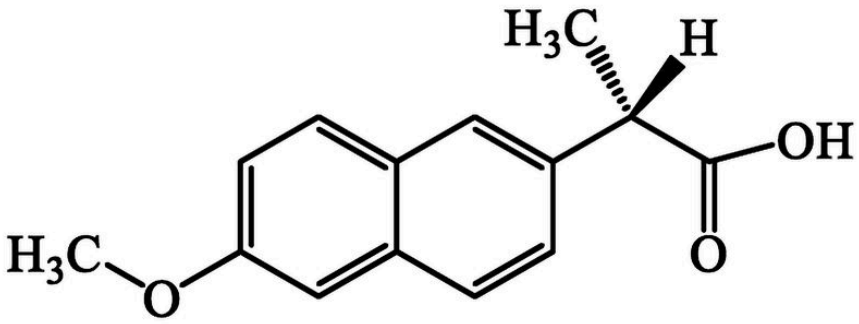
C. By treating the sodium salt with HCl

D. By distillation

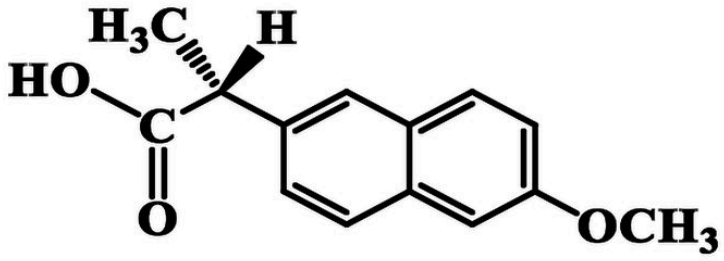
Q59 Which wedge-dash structure is (S)-naproxen?

A. Neither; naproxen has no chiral carbon

B.



C.



Q60 A 50:50 mixture of (+)-glyceraldehyde ([α] = +8.7°) and (−)-glyceraldehyde ([α] = −8.7°) has an observed rotation of:

A. +17.4°

B. −8.7°

C. 0°

D. +8.7°

Q61 The prefix (±) in a name indicates:

A. A racemic mixture

B. A meso compound

C. A structural isomer

D. A pure dextrorotatory enantiomer

3.6 Key Terms & Classification of Isomers

Q62 Stereoisomers are isomers with:

A. Different molecular formulas but the same orientation in space

B. The same connectivity of their atoms but a different orientation in space

C. A different connectivity of their atoms but the same molecular formula

D. The same connectivity and the same orientation of atoms in space

Q63 Structural (constitutional) isomers are isomers with:

A. The same connectivity but a different orientation in space

B. Non-superposable mirror-image molecules

C. Rigidity in their structure

D. A different order of attachment (connectivity) of their atoms

Q64 According to the flow chart, stereoisomers are divided into:

A. Enantiomers and diastereomers

B. Enantiomers and structural isomers

C. Structural and constitutional isomers

D. Conformers and constitutional isomers

Q65 Diastereomers are:

A. Isomers with a different order of attachment of their atoms

B. Stereoisomers whose molecules are not mirror images

C. Stereoisomers whose molecules are non-superposable mirror images

D. Two drawings of the same molecule

Q66 Enantiomers are:

A. Isomers with different molecular formulas

B. Isomers with a different order of attachment of their atoms

C. Stereoisomers whose molecules are non-superposable mirror images

D. Stereoisomers whose molecules are not mirror images

Q67 Geometric isomers are:

A. Stereoisomers whose molecules are non-superposable mirror images

B. Conformations formed by free rotation about single bonds

C. Structural isomers that contain different functional groups

D. Stereoisomers whose molecules possess rigidity in their structure

Q68 A stereogenic center (stereocenter) is:

A. Any carbon atom that is part of a ring

B. A carbon with four different atoms or groups attached (a chiral carbon)

C. A carbon with two identical atoms or groups attached

D. An sp^2 carbon that is part of a C=C double bond

Q69 A chiral molecule is a molecule that:

A. Has an internal plane of symmetry

B. Does not rotate plane-polarized light

C. Can be superposed on its mirror image

D. Cannot be superposed on its mirror image

Q70 A plane of symmetry is:

A. An imaginary plane dividing a molecule into two mirror-image halves

B. The single plane in which plane-polarized light vibrates

C. A plane passing through the four different groups of a chiral carbon

D. The plane of the paper on which a Fischer projection is drawn

Q71 A meso form (meso compound):

A. Has no plane of symmetry, so it rotates plane-polarized light

B. Has only one chiral carbon, so it is optically active

C. Has chiral carbons and an internal plane of symmetry, so it is optically inactive

D. Is a 50:50 mixture of a pair of enantiomers, so it is optically inactive

Q72 Plane-polarized light is:

A. Light of many different colors

B. Light whose waves vibrate in only one plane

C. Light that has passed through a racemic mixture

D. Light whose waves vibrate in all planes

Q73 The specific rotation [α] of an optically active compound is:

A. Its characteristic rotation of plane-polarized light under standard conditions

B. The temperature at which it starts to rotate light

C. The number of chiral carbons present in its molecule

D. The rotation of its racemic mixture, which is always 0°

Q74 Compounds with the same order of bonded atoms that differ only in the direction in which they rotate plane-polarized light are called:

A. Constitutional isomers

B. Conformers

C. Optical isomers (enantiomers)

D. Geometric isomers

Q75 What is the major difference between configuration and conformation?

A. Both are fixed and can only change by breaking bonds

B. Both change freely by rotation about single bonds

C. Conformation is fixed in space; configuration changes by rotation about single bonds

D. Configuration is fixed in space; conformation changes by rotation about single bonds

Answers · Lecture 3

Q1 A	**Q16** B	**Q31** B	**Q46** D	**Q61** A
Q2 D	**Q17** A	**Q32** C	**Q47** D	**Q62** B
Q3 C	**Q18** A	**Q33** A	**Q48** C	**Q63** D
Q4 B	**Q19** B	**Q34** D	**Q49** A	**Q64** A
Q5 A	**Q20** C	**Q35** B	**Q50** B	**Q65** B
Q6 A	**Q21** A	**Q36** B	**Q51** B	**Q66** C
Q7 C	**Q22** D	**Q37** C	**Q52** D	**Q67** D
Q8 B	**Q23** E	**Q38** D	**Q53** A	**Q68** B
Q9 D	**Q24** B	**Q39** A	**Q54** C	**Q69** D
Q10 B	**Q25** A	**Q40** A	**Q55** D	**Q70** A
Q11 D	**Q26** D	**Q41** A	**Q56** A	**Q71** C
Q12 B	**Q27** B	**Q42** B	**Q57** B	**Q72** B
Q13 C	**Q28** C	**Q43** C	**Q58** C	**Q73** A
Q14 A	**Q29** A	**Q44** D	**Q59** B	**Q74** C
Q15 C	**Q30** D	**Q45** C	**Q60** C	**Q75** D

4 Alkanes

In this lecture

1. Hydrocarbons & Classes of Alkanes 5 questions
2. Types of Carbon Atoms 4 questions
3. Physical Properties of Alkanes 2 questions
4. Nomenclature (IUPAC) of Alkanes 4 questions
5. Reactions of Alkanes 1 questions

4.1 Hydrocarbons & Classes of Alkanes

Q1 Which of the following is **NOT** a source of alkanes?

A. Seawater

B. Petroleum

C. Natural gas

D. Coal

Q2 Alkanes can be classified into 3 classes, namely:

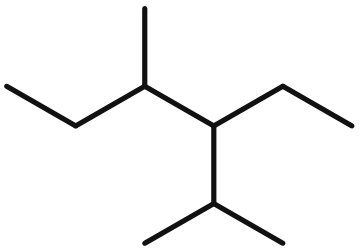
A. Alkenes, alkynes and aromatics

B. Straight-chain, branched-chain and cyclic alkanes

C. Primary, secondary and tertiary alkanes

D. Gases, liquids and solids

Q3 The following compound can be classified as a:



A. Cycloalkane

B. Straight-chain alkane

C. Substituted cycloalkane

D. Branched alkane

Q4 Alkanes are hydrocarbons that have …….:

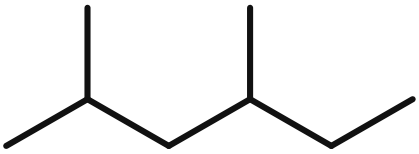
A. Single ω bonds

B. Single α bonds

C. Single σ (sigma) bonds

D. Single π (pi) bonds

Q5 Give the molecular formula for the following compound:



A. C_6H_{14}

B. C_7H_{16}

C. C_8H_{16}

D. C_8H_{18}

4.2 Types of Carbon Atoms

Q6 Carbon atoms present in an alkane can be classified into 4 types of carbons, namely:

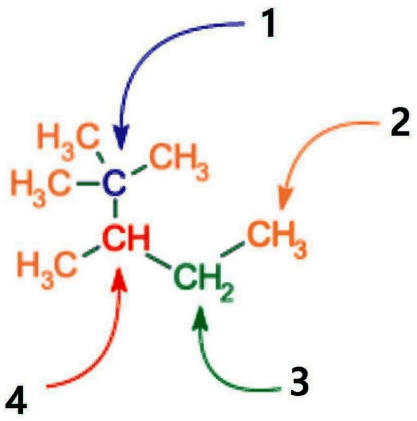
A. sp, sp^2, sp^3 and sp^3d

B. Alpha, beta, gamma and delta

C. Axial, equatorial, cis and trans

D. Primary, secondary, tertiary and quaternary

Q7 In the compound shown, carbon **1** is:



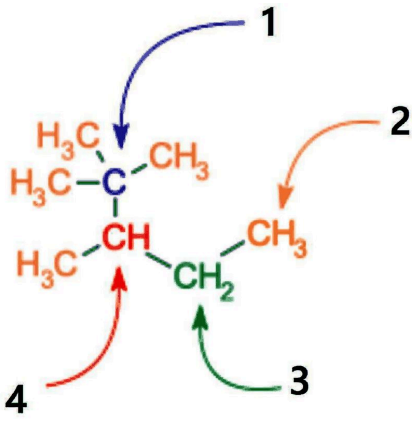
A. Primary (1°)

B. Secondary (2°)

C. Tertiary (3°)

D. Quaternary (4°)

Q8 In the same compound, carbon **4** (the CH) is:



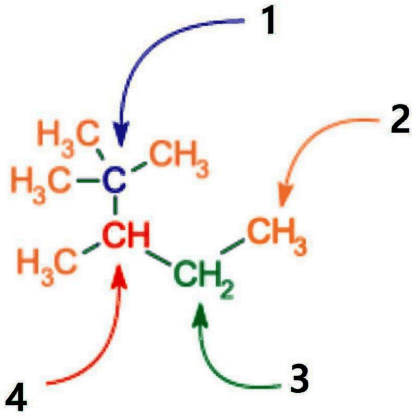
A. Secondary (2°)

B. Primary (1°)

C. Tertiary (3°)

D. Quaternary (4°)

Q9 In the same compound, carbons **2** and **3** are, respectively:



A. Primary (1°) and secondary (2°)

B. Primary (1°) and tertiary (3°)

C. Tertiary (3°) and secondary (2°)

D. Secondary (2°) and primary (1°)

4.3 Physical Properties of Alkanes

Q10 Alkanes are insoluble in water because:

A. They are non-polar and cannot form hydrogen bonds with water

B. They are too heavy

C. They react with water

D. They are ionic compounds

Q11 Which of the following is **NOT** one of the physical properties of alkanes?

A. Lower boiling point than other organic compounds

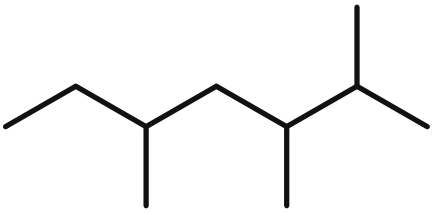
B. Boiling point increases with the molecular weight of the compound

C. Boiling point increases with branching

D. Water-insoluble (insoluble in water)

4.4 Nomenclature (IUPAC) of Alkanes

Q12 What is the name of this alkane (find the longest chain)?



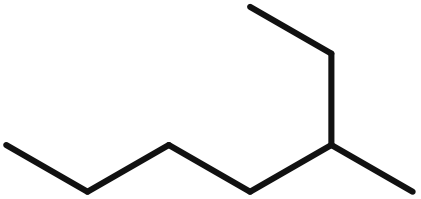
A. 2,3,5-Trimethylheptane

B. 2,3,5-Trimethylhexane

C. 3,5,6-Trimethylheptane

D. Trimethyloctane

Q13 What is the name of this alkane (find the longest chain)?



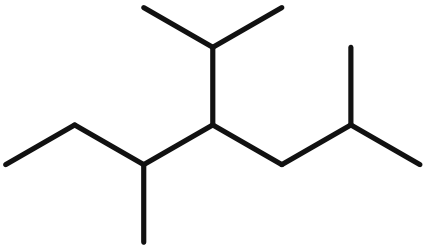
A. 5-Methylheptane

B. 3-Methylheptane

C. 2-Ethylhexane

D. 3-Methylhexane

Q14 Give the correct IUPAC name for this compound:



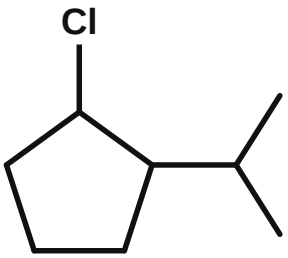
A. 2,5-Dimethyl-4-isopropylheptane

B. 4-Isopropyl-3,6-dimethylheptane

C. 4-Isopropyl-2,5-dimethylheptane

D. 3-Ethyl-4-isopropyl-2-methylpentane

Q15 Give the correct IUPAC name for this compound:



A. 1-Chloro-2-isopropylcyclopentane

B. 1-Isopropyl-2-chlorocyclopentane

C. 2-Chloro-1-isopropylcyclopentane

D. Chloroisopropylpentane

4.5 Reactions of Alkanes

Q16 The 2 chemical reactions of alkanes are:

A. Hydration and hydrolysis

B. Esterification and saponification

C. Acid–base neutralization and reduction

D. Combustion and free-radical halogenation

Answers · Lecture 4

Q1 A	**Q5** D	**Q9** A	**Q13** B
Q2 B	**Q6** D	**Q10** A	**Q14** C
Q3 D	**Q7** D	**Q11** C	**Q15** A
Q4 C	**Q8** C	**Q12** A	**Q16** D

5 Alkenes

In this lecture

1. Structure & Classification of Alkenes 8 questions
2. Nomenclature of Alkenes 6 questions
3. Physical Properties & C–C vs C=C 2 questions
4. Addition Reactions & Markovnikov's Rule 3 questions
5. Oxidation of Alkenes 3 questions
6. Diels–Alder Reaction 2 questions

5.1 Structure & Classification of Alkenes

Q1 What is the general formula for an alkene?

A. C_nH_{2n}

B. C_nH_{2n-2}

C. C_nH_{2n+2}

D. C_nH_n

Q2 Which of the following molecular formulas represents an alkene?

A. C_5H_{12}

B. C_6H_6

C. C_5H_{10}

D. C_6H_8

Q3 In alkenes, the σ bond of the C=C is formed between two _____ hybridized carbons.

A. $sp^2 – sp^3$

B. $sp^3 – sp$

C. $sp – sp^2$

D. $sp^2 – sp^2$

Q4 Which statement about the C=C double bond is correct?

A. It rotates freely like a C–C single bond

B. It is planar with 120° angles, and rotation about it is restricted

C. The π bond is less reactive than the σ bond

D. It is longer than a C–C single bond

Q5 Why is rotation about the C=C double bond restricted?

A. Hydrogen bonding holds the carbons in place

B. The carbons are sp^3 hybridized

C. The σ bond is too strong to rotate

D. The π bond would have to break

Q6 Does isomerism occur in alkenes?

A. Yes — cis/trans (E/Z) geometric isomerism

B. No, never

C. Only in alkenes with triple bonds

D. Only optical isomerism

Q7 Alkenes with more than one double bond are classified into three types:

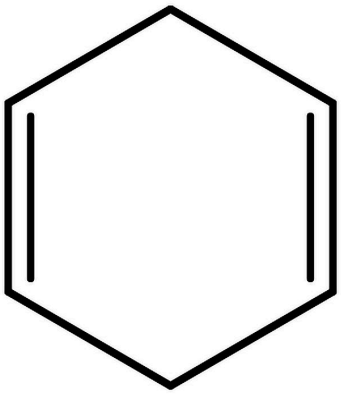
A. Cis, trans and meso

B. Terminal, internal and cyclic

C. Cumulated, conjugated and non-conjugated (isolated)

D. Primary, secondary and tertiary

Q8 The following compound can be classified as:



A. Non-conjugated

B. Cumulated

C. Conjugated

D. Aromatic

5.2 Nomenclature of Alkenes

Q9 Give the IUPAC name of the following compound: $CH_3–CH_2=CH_2–CH_3$

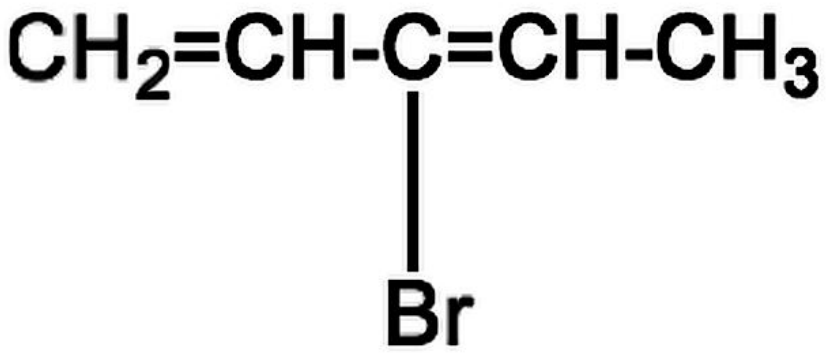
A. 2-Butyne

B. 1-Butene

C. Butane

D. 2-Butene (but-2-ene)

Q10 Give the IUPAC name of the following compound:



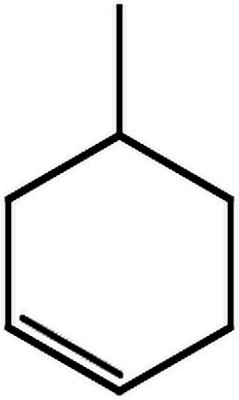
A. 3-Bromo-1-pentene

B. 3-Bromo-2,4-pentadiene

C. 3-Bromo-1,3-pentadiene (3-bromopenta-1,3-diene)

D. 2-Bromo-1,3-pentadiene

Q11 Give the IUPAC name of the following compound:



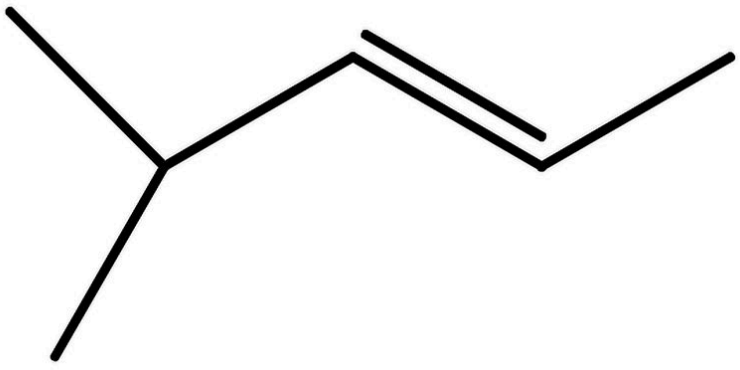
A. 1-Methylcyclohexene

B. 4-Methylcyclohexene (4-methyl-1-cyclohexene)

C. 3-Methylcyclohexene

D. 4-Methylcyclohexane

Q12 Give the IUPAC name of the following compound:



A. 2-Methyl-2-pentene

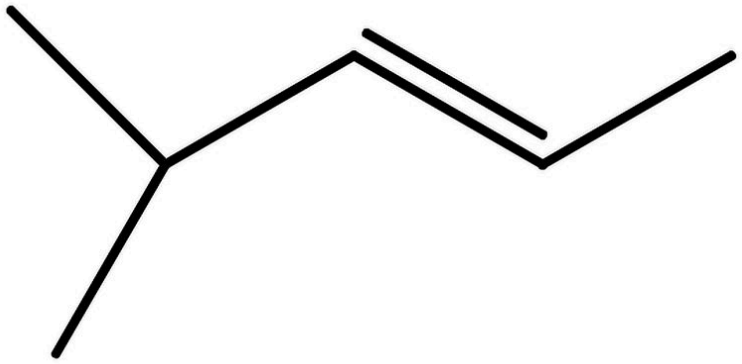
B. 4-Methyl-3-pentene

C. 2-Methyl-3-pentene

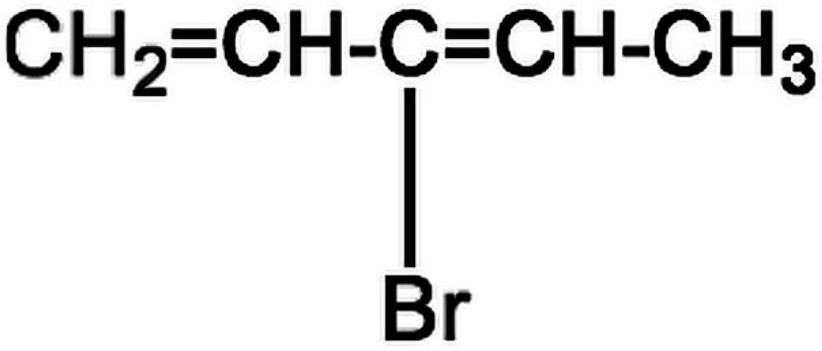
D. 4-Methyl-2-pentene (4-methylpent-2-ene)

Q13 Which structure is 1-bromo-3-heptene?

A.



B.

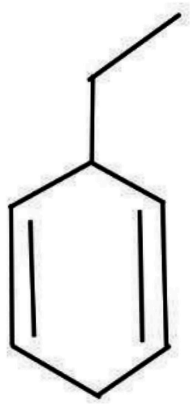


C.

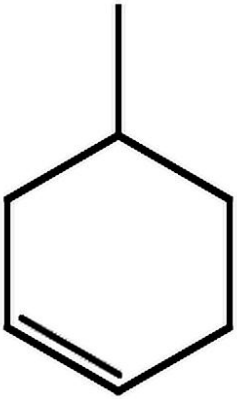


Q14 Which structure is 3-ethyl-1,4-cyclohexadiene?

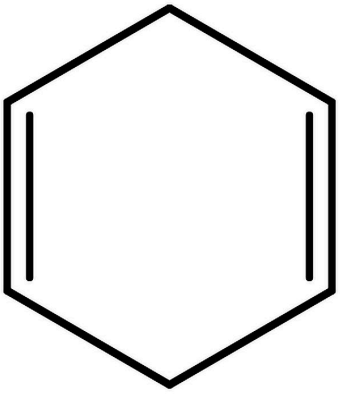
A.



B.



C.



5.3 Physical Properties & C–C vs C=C

Q15 Compare C–C and C=C: which pair is correct?

A. C–C: bond angle 109.5°, 1.54 Å; C=C: bond angle 120°, 1.34 Å

B. Both 109.5°

C. C=C is longer than C–C

D. C–C: 120°, 1.34 Å; C=C: 109.5°, 1.54 Å

Q16 How many atoms are attached to each carbon of a C=C?

A. 2

B. 4

C. 1

D. 3

5.4 Addition Reactions & Markovnikov's Rule

Q17 Which is **NOT** one of the four types of addition reactions of alkenes listed?

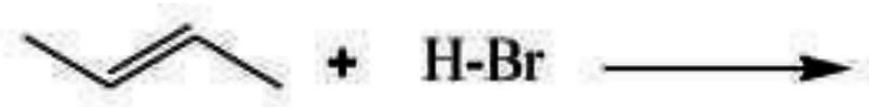
A. Addition of halogens

B. Addition of $NaNH_2$

C. Addition of borane (BH_3)

D. Addition of water (H_2O)

Q18 Complete: 2-butene + HBr →



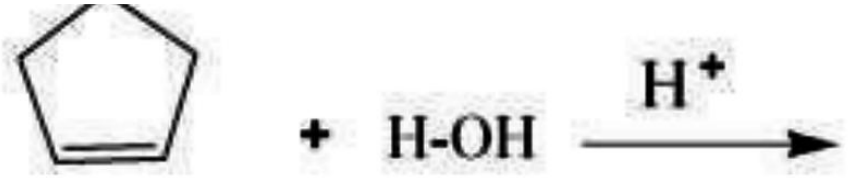
A. Butane

B. 1-Bromobutane

C. 2-Bromobutane

D. 2,3-Dibromobutane

Q19 Complete: cyclopentene + H_2O (H^+) →



A. Cyclopentane-1,2-diol

B. Cyclopentanone

C. Cyclopentene oxide

D. Cyclopentanol

5.5 Oxidation of Alkenes

Q20 Which reagent oxidizes an alkene to an **epoxide**?

A. A peroxycarboxylic acid (3-chloroperbenzoic acid)

B. OsO_4

C. Cold alkaline $KMnO_4$

D. Ozone then Zn/H^+

Q21 Which two reagents oxidize an alkene to a cis-1,2-diol?

A. BH_3 and H_2O_2

B. Cold alkaline $KMnO_4$ and OsO_4

C. Hot $KMnO_4$ and HCl

D. O_3 and H_2O_2

Q22 Ozone (O_3) reacts with alkenes to give which type of products?

A. Carbonyl compounds

B. Alkanes

C. Epoxides

D. Diols

5.6 Diels–Alder Reaction

Q23 The Diels–Alder reaction is:

A. A cycloaddition between a diene and a dienophile giving a cyclic product

B. An elimination reaction

C. An oxidation with $KMnO_4$

D. A free-radical substitution

Q24 The simplest Diels–Alder reaction, 1,3-butadiene + ethylene, gives:

A. Hexane

B. Cyclohexene

C. Cyclobutane

D. Benzene

Answers · Lecture 5

Q1 A	**Q6** A	**Q11** B	**Q16** D	**Q21** B
Q2 C	**Q7** C	**Q12** D	**Q17** B	**Q22** A
Q3 D	**Q8** A	**Q13** C	**Q18** C	**Q23** A
Q4 B	**Q9** D	**Q14** A	**Q19** D	**Q24** B
Q5 D	**Q10** C	**Q15** A	**Q20** A	

6 Alkynes

In this lecture

1. Structure & Classification of Alkynes 3 questions
2. Nomenclature of Alkynes 5 questions
3. Physical Properties & Acidity 2 questions
4. Addition of HX and X_2 3 questions
5. Hydration & Hydroboration 4 questions
6. Acetylide Anions: Alkylation 1 questions
7. Review Questions: Alkenes & Alkynes 10 questions

6.1 Structure & Classification of Alkynes

Q1 What is the general formula for alkynes?

A. C_nH_{2n}

B. C_nH_n

C. C_nH_{2n+2}

D. C_nH_{2n-2}

Q2 Which of the following molecular formulas represents an alkyne?

A. C_6H_6

B. C_5H_8

C. C_5H_{12}

D. C_5H_{10}

Q3 Alkynes can be classified generally into 2 types:

A. Cyclic and aromatic alkynes

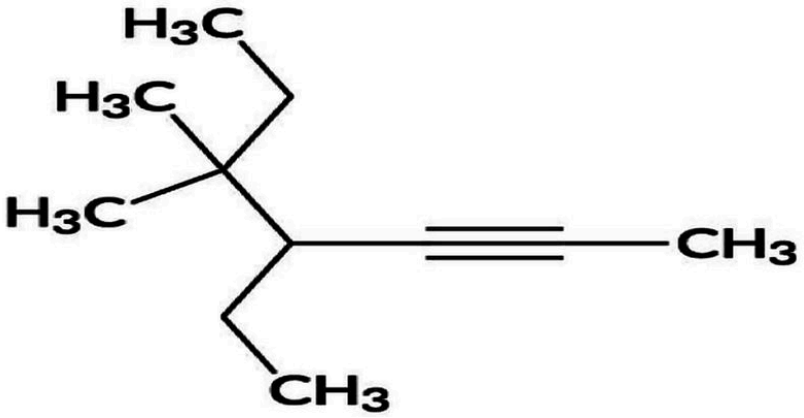
B. Cis and trans alkynes

C. Terminal and internal alkynes

D. Primary and secondary alkynes

6.2 Nomenclature of Alkynes

Q4 Give the IUPAC name of the following compound:



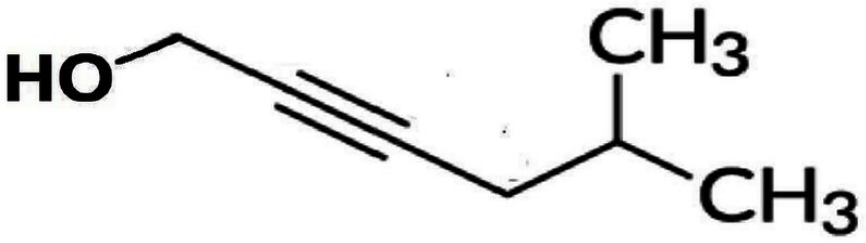
A. 4-Ethyl-3,3-dimethylhept-5-yne

B. 3,3-Dimethyl-4-ethylhept-5-yne

C. 4-Ethyl-5,5-dimethyl-2-hexyne

D. 4-Ethyl-5,5-dimethylhept-2-yne

Q5 Give the IUPAC name of the following compound:



A. 4-Methylhex-2-yn-1-ol

B. 2-Methylhex-4-yn-6-ol

C. 5-Methylhex-2-yn-1-ol

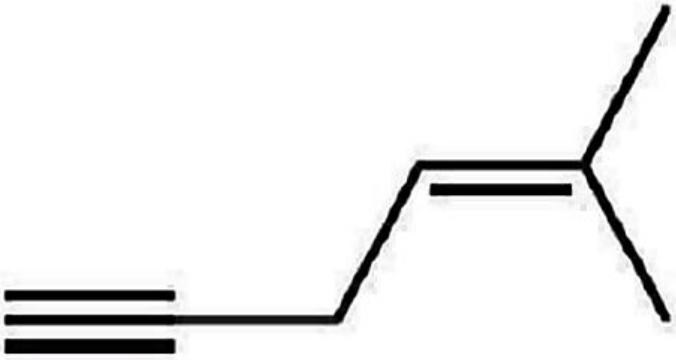
D. 5-Methylhex-3-yn-1-ol

Q6 Which structure is 2,5-dimethylhept-3-yne?

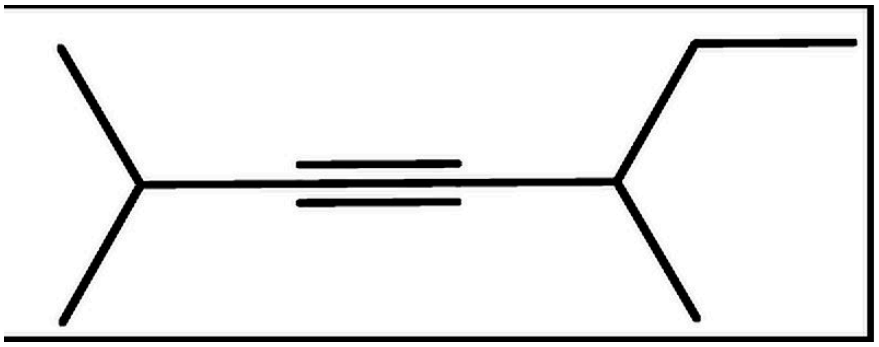
A.



B.

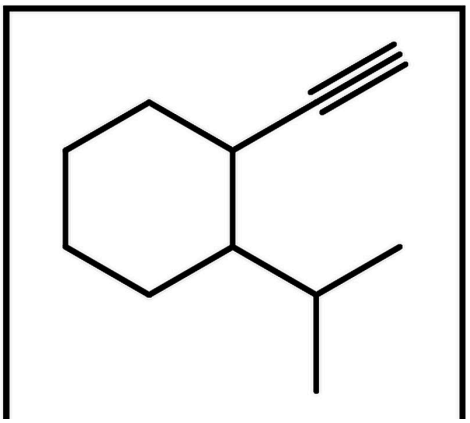


C.

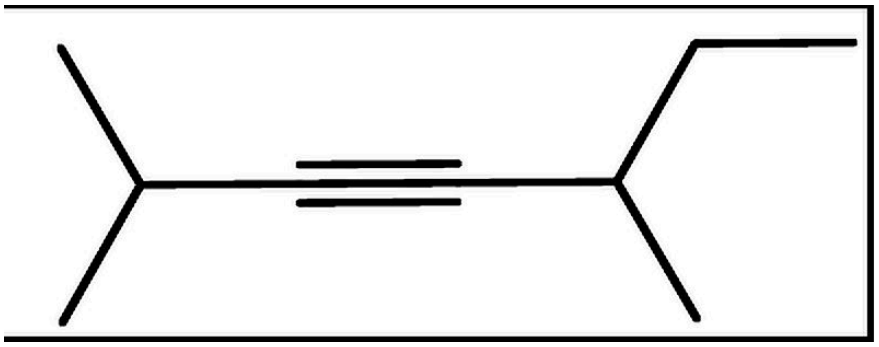


Q7 Which structure is 1-ethynyl-2-isopropylcyclohexane?

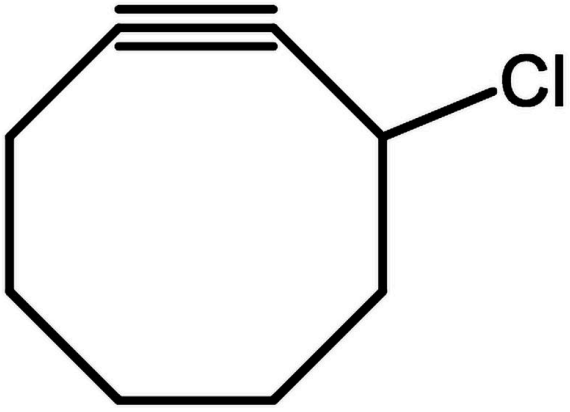
A.



B.

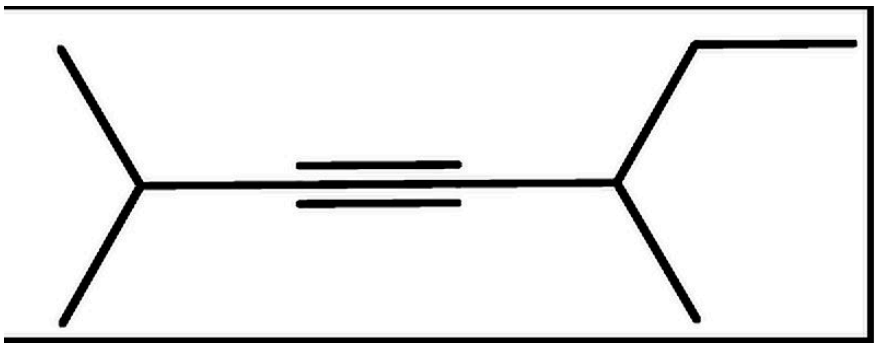


C.

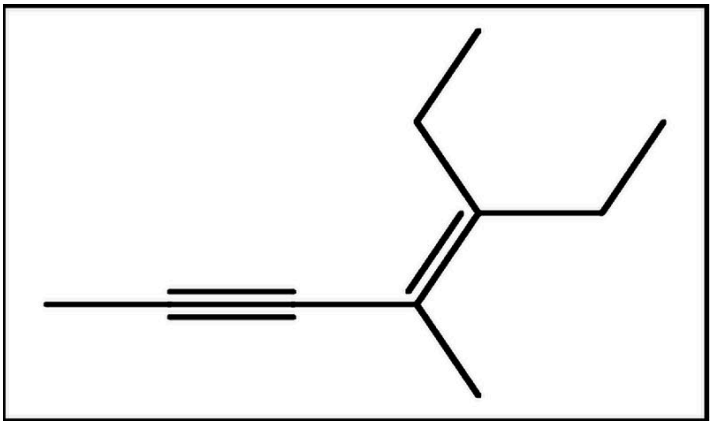


Q8 Which structure is 3-ethyl-4-methylhept-3-en-5-yne?

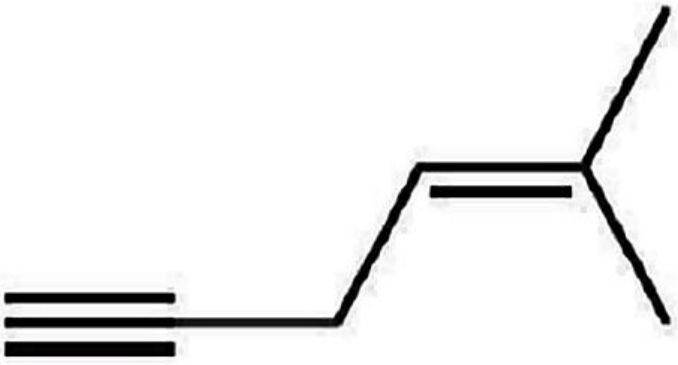
A.



B.



C.



6.3 Physical Properties & Acidity

Q9 Which of the following is **NOT** one of the physical properties of alkynes?

A. Soluble in organic solvents

B. Highly soluble in water

C. Low melting and boiling points

D. Melting and boiling points increase with the number of carbons

Q10 Complete the equation: R–C≡C–H + $NaNH_2$ (liquid NH_3) →

A. A sodium acetylide (R–C≡C:$^-$ Na^+) + NH_3

B. No reaction

C. An alkene + NaH

D. An aldehyde

6.4 Addition of HX and X_2

Q11 Complete the equation: R–C≡C–H + Br_2 (1 equiv.) →

A. A ketone

B. A tetrabromide

C. A geminal dibromide

D. A 1,2-dibromoalkene, R–CBr=CHBr (dihalide)

Q12 Complete the equation: R–C≡C–H + Br_2 (4 equiv.) →

A. An alkyl bromide

B. A 1,2-dibromoalkene

C. No reaction

D. A 1,1,2,2-tetrabromoalkane

Q13 Complete the equation: R–C≡C–H + HCl (1 equiv.) →

A. A geminal dichloride

B. A vinyl chloride with Cl on the more substituted carbon (R–CCl=CH_2)

C. An alkane

D. R–CH=CHCl (Cl on the terminal carbon)

6.5 Hydration & Hydroboration

Q14 Complete the equation: H–C≡C–CH_3 + H_2O (H_2SO_4, $HgSO_4$) →

A. An alkane

B. A methyl ketone, acetone (via an enol)

C. A primary alcohol

D. An aldehyde

Q15 Complete the equation: R–C≡C–CH_3 + H_2O (H_2SO_4) →

A. A carboxylic acid

B. A ketone

C. An ether

D. An aldehyde

Q16 Complete the equation: Ph–C≡C–H + 1) THF, BH_3 2) H_2O_2, OH^- →

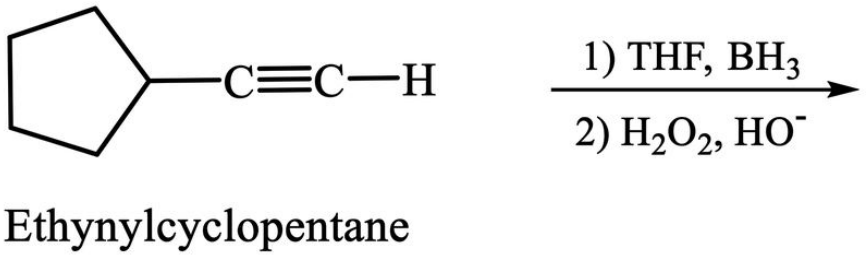
A. Acetophenone (Ph–CO–CH_3)

B. Benzoic acid

C. Styrene

D. Phenylacetaldehyde (Ph–CH_2–CHO)

Q17 The reaction of ethynylcyclopentane with BH_3 in THF, followed by basic H_2O_2/OH^-, gives:



Ethynylcyclopentane

A. Cyclopentanecarboxylic acid

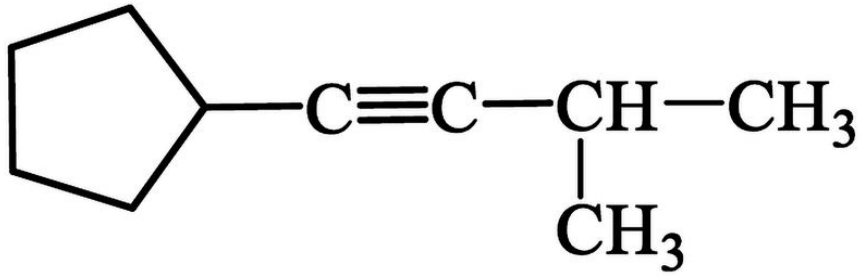
B. 2-Cyclopentylethanol

C. 1-Cyclopentylethanone (cyclopentyl–CO–CH_3)

D. Cyclopentylacetaldehyde (cyclopentyl–CH_2–CHO)

6.6 Acetylide Anions: Alkylation

Q18 Suggest starting materials and reagents for the alkyne shown:



A. Ethynylcyclopentane + $NaNH_2$, then 2-bromopropane (acetylide alkylation)

B. Cyclopentene + Br_2

C. Ethynylcyclopentane + $H_2O/HgSO_4$

D. Cyclopentanol + H_2SO_4

6.7 Review Questions: Alkenes & Alkynes

Q19 What is the major organic product of the acid-catalyzed hydration of 1-methylcyclohexene?

A. 1-Methylcyclohexene oxide

B. Cyclohexylmethanol

C. 2-Methylcyclohexanol

D. 1-Methylcyclohexanol

Q20 The addition of water to an alkene in the presence of a strong acid catalyst is called:

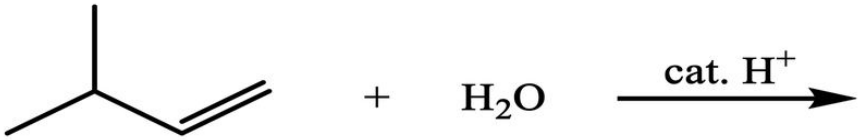
A. Acid-catalyzed hydration

B. Hydrogenation

C. Hydroboration

D. Dehydration

Q21 Predict the major product of this reaction:



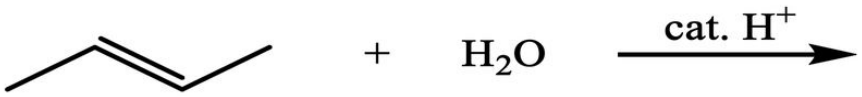
A. 3-Methylbutan-2-ol

B. 2-Methylbutan-1-ol

C. 3-Methylbutan-1-ol

D. 2-Methylbutan-2-ol

Q22 Predict the major product of this reaction:



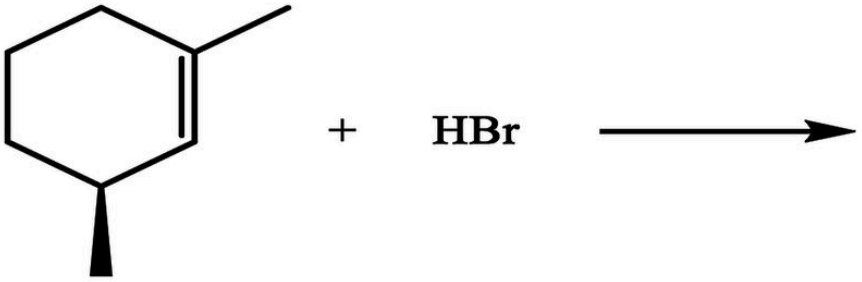
A. Butan-2-ol

B. Butan-1-ol

C. Butane

D. 2-Methylpropan-2-ol

Q23 Predict the major product of this reaction:



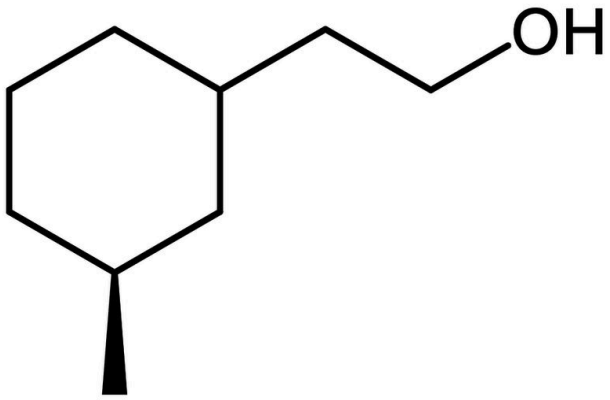
A. 1,3-Dimethylcyclohexane

B. 1-Bromo-1,3-dimethylcyclohexane

C. 2-Bromo-1,3-dimethylcyclohexane

D. 1-Bromo-2,4-dimethylcyclohexane

Q24 Which alkene and reagents are needed to prepare this alcohol?



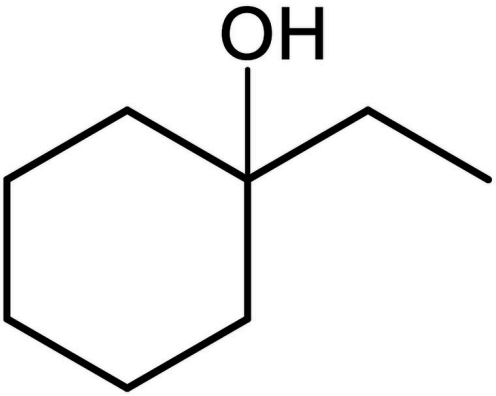
A. Ethylidene-3-methylcyclohexane + HBr

B. 3-Methylcyclohexene + H_2O, H_2SO_4

C. 1-Methyl-3-vinylcyclohexane, then 1) BH_3/THF 2) H_2O_2, OH^-

D. 1-Methyl-3-vinylcyclohexane + H_2O, H_2SO_4

Q25 Which alkene and reagents are needed to prepare 1-ethylcyclohexanol?



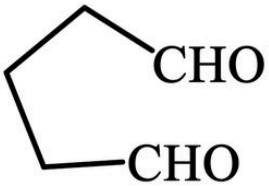
A. Vinylcyclohexane, then 1) BH_3 2) H_2O_2, OH^-

B. 1-Ethylcyclohexene + H_2O, H_2SO_4 (acid-catalyzed hydration)

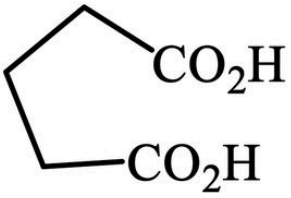
C. Cyclohexene + ethanol

D. 1-Ethylcyclohexene, then 1) BH_3 2) H_2O_2, OH^-

Q26 Starting with cyclopentene, which reagents give pentanedial (i) and pentanedioic acid (ii)?



(i)



(ii)

A. (i) HBr; (ii) Br_2

B. (i) H_2O, H_2SO_4; (ii) BH_3 then H_2O_2

C. (i) H_2, Pd; (ii) $NaNH_2$

D. (i) 1) O_3 2) Zn/H_2O (or $(CH_3)_2S$); (ii) hot $KMnO_4$ (or O_3 then H_2O_2)

Q27 The reaction of 1-methylcyclopentene with BH_3 in THF, followed by basic H_2O_2/OH^-, gives:

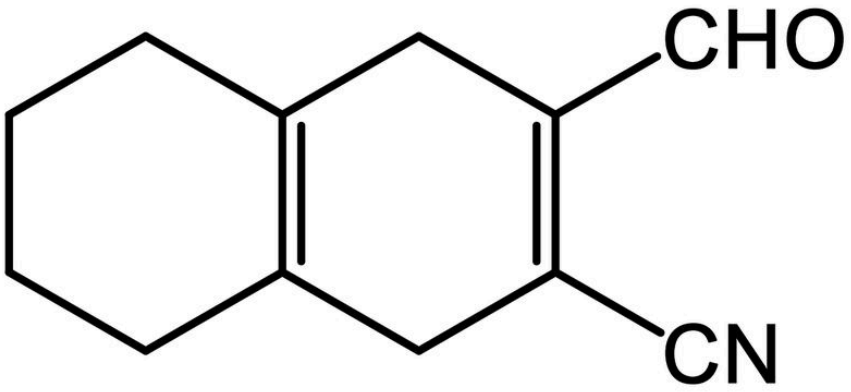
A. 1-Methylcyclopentan-1-ol

B. cis-2-Methylcyclopentan-1-ol

C. trans-2-Methylcyclopentan-1-ol

D. Cyclopentylmethanol

Q28 Which diene and dienophile react (Diels–Alder) to give the bicyclic molecule shown?



A. 1,2-Dimethylenecyclohexane + NC–C≡C–CHO

B. Buta-1,3-diene + cyclohexyne

C. 1,3-Cyclohexadiene + acrylonitrile

D. Cyclohexene + NC–CH=CH–CHO

Answers · Lecture 6

Q1 D	**Q7** A	**Q13** B	**Q19** D	**Q25** B
Q2 B	**Q8** B	**Q14** B	**Q20** A	**Q26** D
Q3 C	**Q9** B	**Q15** B	**Q21** D	**Q27** C
Q4 D	**Q10** A	**Q16** D	**Q22** A	**Q28** A
Q5 C	**Q11** D	**Q17** D	**Q23** B	
Q6 C	**Q12** D	**Q18** A	**Q24** C	

7 Ethers & Epoxides

In this lecture

1. Ethers: Structure & Nomenclature 8 questions
2. Physical Properties of Ethers 3 questions
3. Preparation of Ethers 5 questions
4. Reactions of Ethers with HX 10 questions
5. Epoxides: Structure & Preparation 5 questions
6. Ring-Opening Reactions of Epoxides 9 questions

7.1 Ethers: Structure & Nomenclature

Q1 The general structure of an ether is:

A. R–CO–R′

B. R–COOH

C. R–OH

D. R–O–R′

Q2 In common nomenclature, ethers are named by:

A. Naming the alkyl groups by size, largest first, followed by the word “oxide”

B. Naming the alkyl groups on oxygen in alphabetical order, followed by the word “ether”

C. Naming the longest chain as the parent and adding the suffix -ol to it

D. Naming the longer alkyl group as an alkoxy substituent on the shorter one

Q3 In IUPAC nomenclature of ethers, the –OR group is named as:

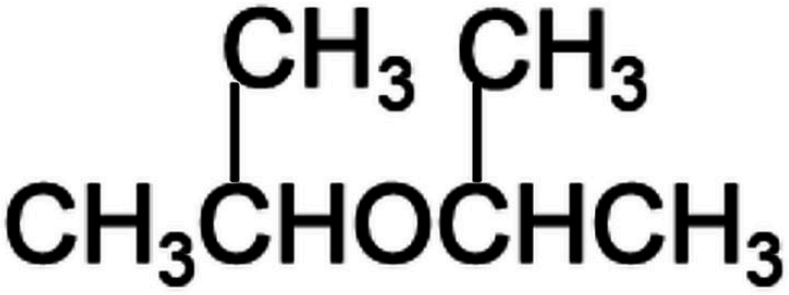
A. An alkoxy substituent on the longest carbon chain

B. An -ol suffix added to the longest chain

C. A suffix “-ether” added to the longest chain

D. The parent chain, with the longer alkyl group as a substituent

Q4 What is the common name of this ether?



A. Isopropyl propyl ether

B. Dimethyl ether

C. Diisopropyl ether

D. Dipropyl ether

Q5 What is the common name of $CH_3CH_2OCH_3$?



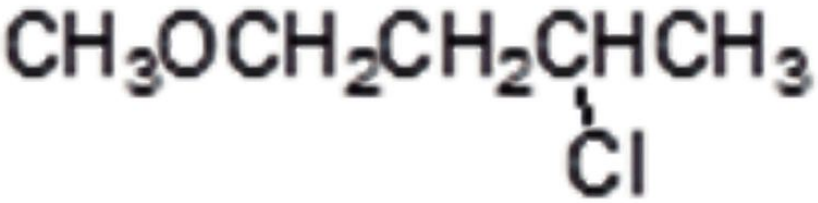
A. Methoxyethene

B. Diethyl ether

C. Ethyl propyl ether

D. Ethyl methyl ether

Q6 What is the IUPAC name of this compound?



A. 3-Chloro-1-methoxybutane

B. 2-Chloro-4-methoxybutane

C. 2-Chloro-4-methoxypentane

D. 1-Methoxy-3-chlorobutane

Q7 What is the IUPAC name of $CH_3OCH_2CH_2CH_2OH$?



A. 1-Methoxypropane

B. 1-Methoxy-3-propanol

C. 3-Methoxy-1-propanol

D. 1-Hydroxy-3-methoxypropane

Q8 What is the IUPAC name of $CH_3CH_2CH_2OCH_2(CH_2)_5CH_3$?



A. 1-Heptoxypropane

B. 1-Propoxyheptane

C. 1-Propoxyoctane

D. 2-Propoxyheptane

7.2 Physical Properties of Ethers

Q9 Using the table, why does 1-butanol (118 °C) boil much higher than diethyl ether (35 °C)?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
1-butanol $CH_3CH_2CH_2CH_2CH_2OH$	118	74	7.9
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5

A. 1-Butanol is non-polar, so its molecules pack more tightly

B. Ether molecules form stronger hydrogen bonds with each other

C. Diethyl ether has a much higher molecular weight than 1-butanol

D. Alcohol molecules form hydrogen bonds with each other; ether molecules cannot

Q10 Why does diethyl ether have a boiling point similar to pentane?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5
pentane $CH_3CH_2CH_2CH_2CH_3$	36	72	0.03

A. There is no hydrogen bonding between ether molecules

B. Ether molecules form strong hydrogen bonds with each other

C. Both compounds are ionic and have similar charges

D. Both compounds contain an OH group in their molecules

Q11 Why is diethyl ether much more soluble in water (7.5 g/100 mL) than pentane (0.03 g/100 mL)?

Compound	BP (°C)	MW	Water sol. (g/100 mL, 20 °C)
Diethyl ether $CH_3CH_2OCH_2CH_3$	35	74	7.5
pentane $CH_3CH_2CH_2CH_2CH_3$	36	72	0.03

A. Ethers can donate H-bonds to water through an O–H group

B. Ethers react with water to form water-soluble alcohols

C. Pentane is more polar than ether, so it repels water

D. Ethers can accept H-bonds from water through the oxygen lone pairs

7.3 Preparation of Ethers

Q12 The two methods used to prepare ethers are:

A. Ozonolysis of alkenes and hydroboration of alkynes

B. Air oxidation of alkenes and peroxyacid oxidation

C. Williamson synthesis and acid-catalyzed dehydration of alcohols

D. Combustion of alkanes and free-radical halogenation

Q13 In the Williamson synthesis, the alkyl halide should be:

A. Methyl, primary or benzylic

B. An aryl halide

C. Tertiary only

D. Secondary or tertiary

Q14 In the Williamson synthesis (R–X + NaOR′), the R′ group of the alkoxide can be:

A. 2° or 3° only

B. CH_3, 1°, 2°, 3° or aryl

C. CH_3, 1° or benzylic only

D. Aryl only

Q15 In the doctor's Q7 table, the dehydration-of-alcohols row gives the reagent H_2SO_4 and the product $CH_3CH_2OCH_2CH_3$ (a symmetrical ether). The starting material is:

A. Methanol (CH_3OH)

B. Bromoethane (CH_3CH_2Br)

C. Ethanol (CH_3CH_2OH)

D. Ethene ($CH_2{=}CH_2$)

Q16 To make an unsymmetrical ether from $CH_3CH_2CH_2Br$ by the Williamson synthesis, a suitable reagent and product are:

A. $H_2SO_4 \rightarrow CH_3CH_2CH_2OCH_2CH_2CH_3$

B. $NaOH \rightarrow CH_3CH_2CH_2OH$

C. $HBr \rightarrow CH_3CH_2CH_2OCH_3$

D. $NaOCH_3 \rightarrow CH_3CH_2CH_2OCH_3$

7.4 Reactions of Ethers with HX

Q17 The reaction sites in an ether R–O–R are:

A. A polar C=O double bond and the O–H bond

B. One polar O–H bond and one C–O bond

C. Only the C–H bonds of the two alkyl groups

D. Two polar C–O bonds and two non-bonding electron pairs on O

Q18 An ether reacts with **one** mole of HX (X = Cl, Br, I) to give:

A. R–X + R–OH

B. No reaction

C. R–OH + R–OH

D. 2 R–X + H_2O

Q19 An ether reacts with **two** moles of HX (X = Cl, Br, I) to give:

A. 2 R–OH

B. R–X + R–OH

C. 2 R–X + H_2O

D. R–O–X + R–H

Q20 When an ether is cleaved by HX, the first step is:

A. Loss of a carbocation from the ether

B. Formation of an alkyl radical

C. Attack of X^- on the ether oxygen

D. Protonation of the ether oxygen by H^+

Q21 Ether cleavage by HX follows SN1 when the carbon attached to oxygen is:

A. Primary or methyl

B. Secondary or tertiary

C. Aromatic (aryl)

D. Any carbon, whatever its type

Q22 Ether cleavage by HX through the SN2 mechanism occurs at a:

A. Tertiary (3°) carbon, in a 2-step process

B. Secondary (2°) carbon, in a 2-step process

C. Primary (1°) carbon, in a 1-step process

D. Tertiary (3°) carbon, in a 1-step process

Q23 Complete: $CH_3OC(CH_3)_3$ + HBr →

A. CH_3OH + $(CH_3)_3CBr$, by SN1

B. CH_3Br + $(CH_3)_3COH$, by SN2

C. CH_3Br + $(CH_3)_3CBr$ + H_2O

D. No reaction

Q24 Complete: $CH_3OC_2H_5$ + 2 HI →

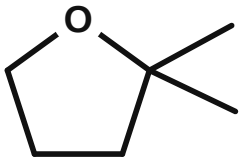
A. C_2H_5I + CH_3OH, by SN2

B. CH_3I + C_2H_5OH, by SN1

C. CH_3I + C_2H_5I + H_2O, by SN2

D. CH_3I + C_2H_5I + H_2O, by SN1

Q25 Complete the reaction of this cyclic ether with HCl:



A. HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, by SN1

B. HO–$CH_2CH_2CH_2$–$C(CH_3)_2Cl$, by SN2

C. Cl–$CH_2CH_2CH_2$–$C(CH_3)_2OH$, by SN2

D. Cl–$CH_2CH_2CH_2$–$C(CH_3)_2OH$, by SN1

Q26 Hint from the doctor: the halide attacks the R group that gives the more stable carbocation. Which order of carbocation stability is correct?

A. 1° > 2° > 3° > CH_3

B. 3° > 2° > 1° > CH_3

C. 2° > 3° > 1° > CH_3

D. CH_3 > 1° > 2° > 3°

7.5 Epoxides: Structure & Preparation

Q27 Epoxides are:

A. Six-membered cyclic ethers containing one oxygen atom

B. Five-membered cyclic ethers containing two oxygen atoms

C. Three-membered cyclic ethers containing one oxygen atom

D. Three-membered cyclic alcohols containing one OH group

Q28 The simplest epoxide is:

A. Tetrahydrofuran

B. Ethylene oxide

C. Diethyl ether

D. Dioxane

Q29 How do epoxides differ from other cyclic ethers (such as those shown below) in structure?



A. Other cyclic ethers have rings larger than the three-membered epoxide ring

B. Epoxides contain two oxygen atoms in the ring

C. Epoxides have a larger ring than other cyclic ethers

D. Other cyclic ethers have no oxygen in the ring

Q30 Ethylene oxide is made industrially from ethene by:

A. Addition of HBr followed by heating

B. Hydroboration followed by oxidation

C. Williamson synthesis with sodium ethoxide

D. Air oxidation with O_2 over a silver catalyst

Q31 Cyclohexene reacts with a peroxycarboxylic acid (RCO_3H) to give:

A. Cyclohexane-1,2-diol only

B. Cyclohexanol + RCOOH

C. Cyclohexane + RCO_3H

D. Cyclohexene oxide + RCOOH

7.6 Ring-Opening Reactions of Epoxides

Q32 Why are epoxides more reactive than ordinary ethers?

A. Their bond angles are widened from ~109° to ~120°

B. Their bond angles are distorted from ~109° to ~60°, causing ring strain

C. Their oxygen has no lone pairs, so it is very electron-poor

D. They contain a C=C double bond next to the oxygen

Q33 In acid-catalyzed ring opening of ethylene oxide with water, what happens in the **last** step of the mechanism?

A. Nucleophilic attack of H_2O on a ring carbon

B. Loss of a water molecule, giving an alkene

C. Loss of a hydrogen ion, giving the 1,2-diol (ethylene glycol)

D. Protonation of the epoxide oxygen

Q34 In acid-catalyzed ring opening of an epoxide with water, what is the correct order of the mechanism steps?

A. Protonation of the epoxide O → nucleophilic attack of H_2O → loss of a hydrogen ion

B. Protonation of the epoxide O → loss of a hydrogen ion → nucleophilic attack of H_2O

C. Nucleophilic attack of H_2O → protonation of the epoxide O → loss of a hydrogen ion

D. Loss of a hydrogen ion → protonation of the epoxide O → nucleophilic attack of H_2O

Q35 Acid-catalyzed hydrolysis of ethylene oxide gives:

A. Diethyl ether ($CH_3CH_2OCH_2CH_3$)

B. Ethylene glycol ($HOCH_2CH_2OH$)

C. Ethanol (CH_3CH_2OH)

D. Acetaldehyde (CH_3CHO)

Q36 Complete the reaction:



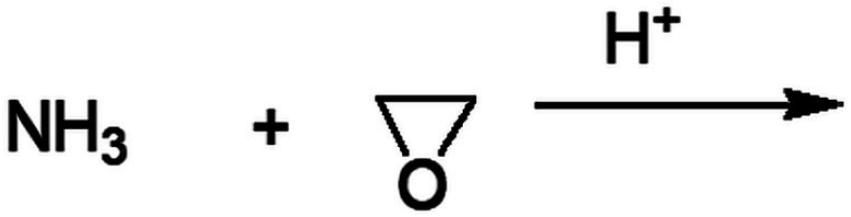
A. $HOCH_2CH_2OH$

B. $CH_3OCH_2CH_2OCH_3$

C. $CH_3OCH_2CH_3$

D. $HOCH_2CH_2OCH_3$

Q37 Complete the reaction:



A. $H_2NCH_2CH_2OH$

B. $HOCH_2CH_2OH$

C. $H_2NCH_2CH_2NH_2$

D. $CH_3CH_2NH_2$

Q38 A Grignard reagent (R–MgX) reacts with ethylene oxide, followed by water, to give:

A. $R–O–CH_2CH_3$

B. RCH_2CH_2OR

C. RCH_2CH_2OH

D. RCH_2OH

Q39 An organolithium reagent (R–Li) reacts with ethylene oxide. Before water is added, the intermediate formed is:

A. $R–O–CH_2CH_2Li$ (a lithium alkyl)

B. RCH_2CH_2OLi (a lithium alkoxide)

C. RCH_2CH_2Li (a lithium alkyl)

D. RCH_2OLi (a lithium alkoxide)

Q40 Epoxides undergo ring-opening reactions with which nucleophiles (HA)?

A. O_2, O_3, N_2, CO_2

B. Alkenes, alkynes, H_2, CH_4

C. H_2O, ROH, R_2NH, HX

D. H_2, CH_4, C_2H_6, N_2

Answers · Lecture 7

Q1 D	**Q9** D	**Q17** D	**Q25** A	**Q33** C
Q2 B	**Q10** A	**Q18** A	**Q26** B	**Q34** A
Q3 A	**Q11** D	**Q19** C	**Q27** C	**Q35** B
Q4 C	**Q12** C	**Q20** D	**Q28** B	**Q36** D
Q5 D	**Q13** A	**Q21** B	**Q29** A	**Q37** A
Q6 A	**Q14** B	**Q22** C	**Q30** D	**Q38** C
Q7 C	**Q15** C	**Q23** A	**Q31** D	**Q39** B
Q8 B	**Q16** D	**Q24** C	**Q32** B	**Q40** C

8 Alcohols

In this lecture

1. Structure & Classification of Alcohols 11 questions
2. Nomenclature of Alcohols 13 questions
3. Physical Properties of Alcohols 5 questions
4. Preparation of Alcohols 14 questions
5. Reactions of Alcohols 26 questions

8.1 Structure & Classification of Alcohols

Q1 An alcohol (ROH) is:

A. An alkane in which one CH_2 is replaced by an O atom

B. An alkane in which one hydrogen is replaced by an OH group

C. An alkane in which one CH_2 is replaced by a C=O group

D. An alkane in which one CH_3 is replaced by a COOH group

Q2 Alcohols are classified as 1°, 2° or 3° according to:

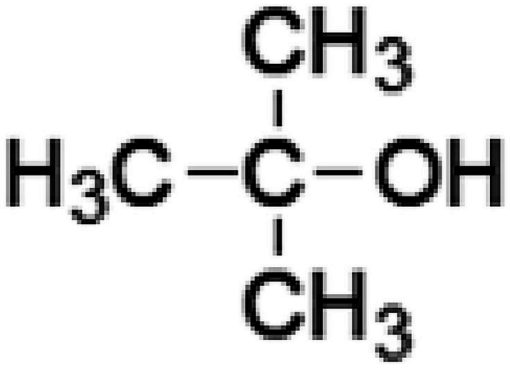
A. The number of OH groups attached to the carbon chain

B. The number of carbon atoms in the longest chain

C. The number of alkyl (R) groups bonded directly to the carbon carrying the OH

D. The number of hydrogen atoms bonded to the carbon carrying the OH

Q3 Classify this alcohol:



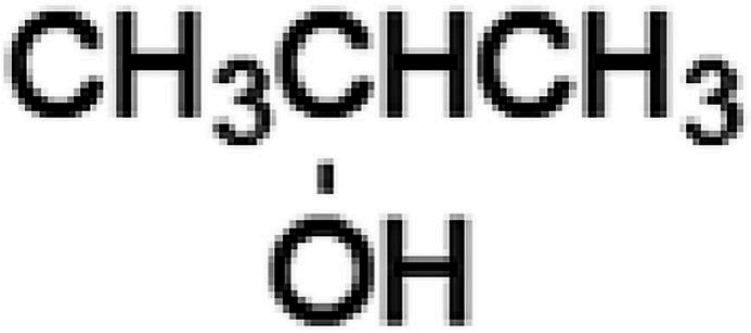
A. Tertiary (3°)

B. Primary (1°)

C. Secondary (2°)

D. Quaternary

Q4 Classify this alcohol:



A. Quaternary

B. Tertiary (3°)

C. Primary (1°)

D. Secondary (2°)

Q5 Classify this alcohol:



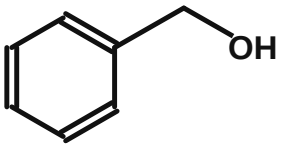
A. Secondary (2°)

B. Phenol

C. Tertiary (3°)

D. Primary (1°)

Q6 Classify this alcohol:



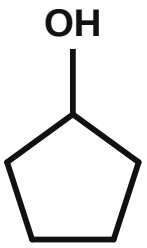
A. Primary (1°)

B. A phenol

C. Tertiary (3°)

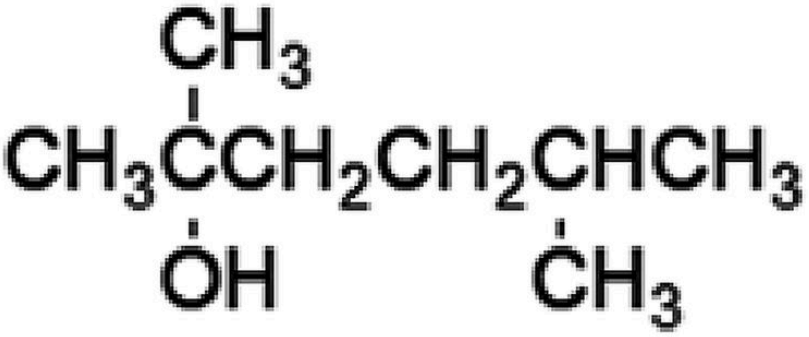
D. Secondary (2°)

Q7 Classify this alcohol:



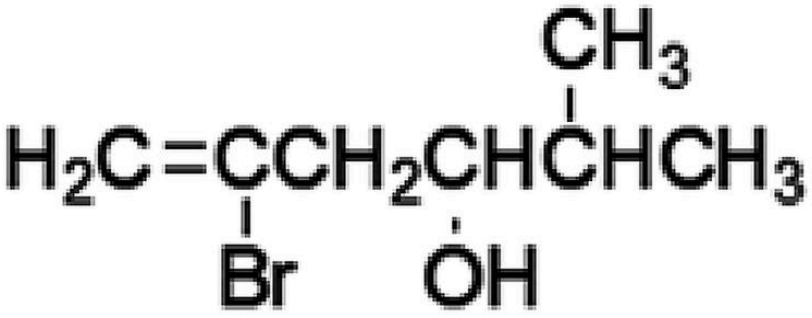
A. Primary (1°)
B. Quaternary
C. Secondary (2°)
D. Tertiary (3°)

Q8 Classify this alcohol:



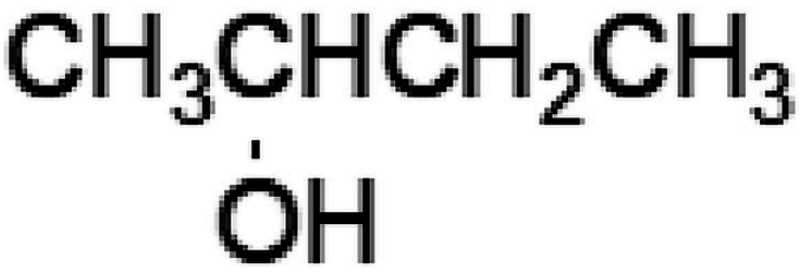
A. Primary (1°)
B. Tertiary (3°)
C. Quaternary
D. Secondary (2°)

Q9 Classify this alcohol:



A. Secondary (2°)
B. Quaternary
C. Primary (1°)
D. Tertiary (3°)

Q10 Classify this alcohol:



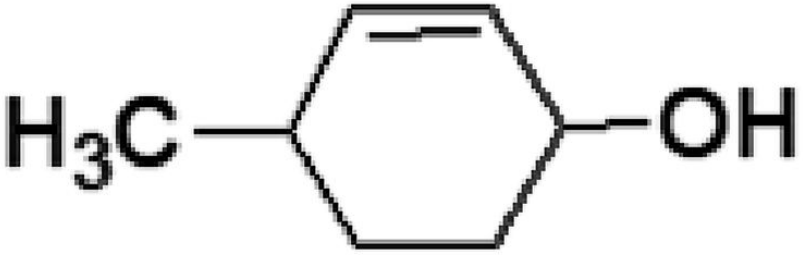
A. Primary (1°)

B. Secondary (2°)

C. Quaternary

D. Tertiary (3°)

Q11 Classify this alcohol:



A. Tertiary (3°)

B. Quaternary

C. Secondary (2°)

D. Primary (1°)

8.2 Nomenclature of Alcohols

Q12 In IUPAC naming of alcohols, the parent chain is:

A. The chain with the most branches attached to it

B. The longest chain, even if it does not contain the OH

C. The shortest chain that contains the OH group

D. The longest continuous carbon chain containing the OH group

Q13 When numbering an alcohol chain, the lowest number goes first to:

A. The triple bond

B. The halogen

C. The carbon carrying the OH group

D. The double bond

Q14 What is the IUPAC name of this compound?



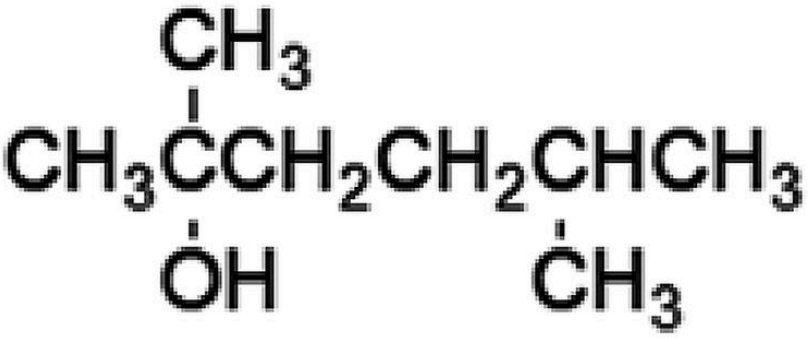
A. 2-Bromo-3-hexyn-6-ol

B. 5-Bromo-3-hexyn-1-ol

C. 5-Bromo-3-hexen-1-ol

D. 2-Bromo-3-hexen-6-ol

Q15 What is the IUPAC name of this compound?



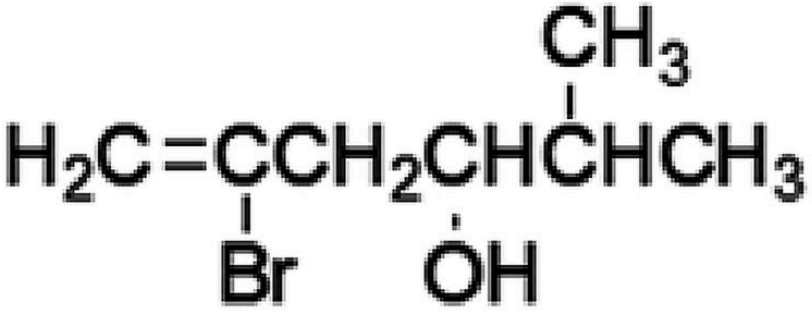
A. 2,5-Dimethylhexan-2-ol

B. 2-Methyl-5-methylhexan-2-ol

C. Octan-2-ol

D. 2,5-Dimethylhexan-5-ol

Q16 What is the IUPAC name of this compound?



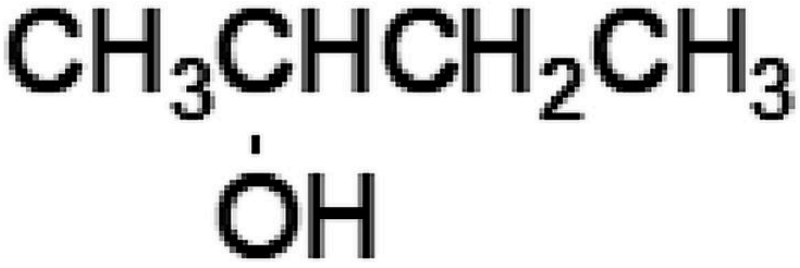
A. 2-Bromo-5-methylhex-1-ene

B. 2-Bromo-5-methyl-1-hexen-4-ol

C. 5-Bromo-2-methylhexan-3-ol

D. 5-Bromo-2-methyl-5-hexen-3-ol

Q17 What is the IUPAC name of $CH_3CH(OH)CH_2CH_3$?



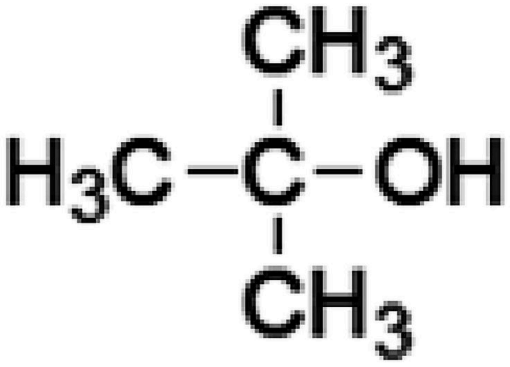
A. 2-Butanone

B. 2-Butanol

C. 1-Butanol

D. 1-Butanone

Q18 What is the IUPAC name of this compound?



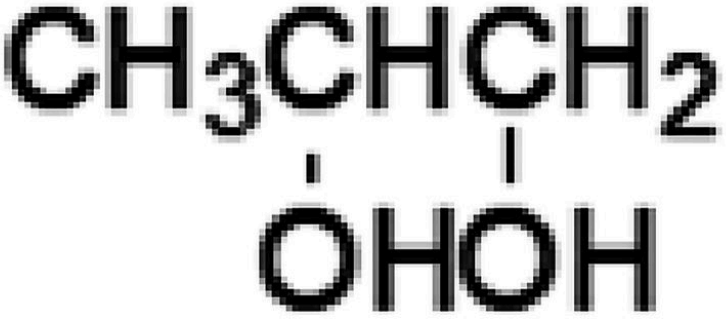
A. 2-Methyl-1-propanol

B. 2,2-Dimethylethanol

C. 2-Methyl-2-propanol

D. 2-Butanol

Q19 What is the IUPAC name of this compound?



A. 1,2-Propanediol

B. 2,3-Propanediol

C. 1,2-Propanetriol

D. 1,2-Propanol

Q20 What is the IUPAC name of this compound?



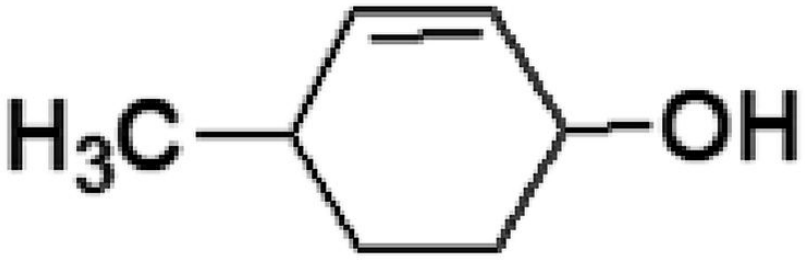
A. 1,2,3-Propanol

B. 1,2,3-Propanetriol

C. 2-Propanol

D. 1,3-Propanediol

Q21 What is the IUPAC name of this compound?



A. 4-Methyl-2-cyclohexen-1-ol

B. 4-Methylcyclohexanol

C. 4-Methyl-3-cyclohexen-1-ol

D. 1-Methyl-3-cyclohexen-4-ol

Q22 What is the IUPAC name of this compound?



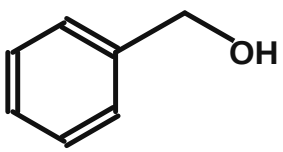
A. 2-Propanone

B. Propan-1,2-ol

C. 1-Propanol

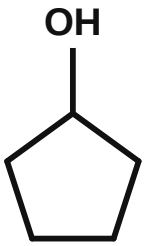
D. 2-Propanol

Q23 What is the IUPAC name of this compound?



A. Methylphenol
B. Phenylethanol
C. Phenylmethanol
D. Phenol

Q24 What is the IUPAC name of this compound?



A. Cyclopentanone
B. Cyclohexanone
C. Cyclopentanol
D. Cyclohexanol

8.3 Physical Properties of Alcohols

Q25 Which molecule has the highest boiling point?

A. $CH_3(CH_2)_5CH_2OH$
B. $CH_3OCH_2CH_3$
C. $CH_3CH_2CH_3$
D. $CH_3CH_2CH_2OH$

Q26 What can be concluded from the table?

	mwt	bp °C
CH_3OH	32	65
CH_3CH_2OH	46	78.5
$CH_3CH_2CH_2OH$	60	97
$CH_3(CH_2)_3CH_2OH$	80*	137
$CH_3(CH_2)_5CH_2OH$	116	176

*As written in the doctor's table.

A. Boiling point stays the same, whatever the molecular weight

B. Boiling point decreases with increasing molecular weight within a homologous series

C. Boiling point increases with increasing molecular weight within a homologous series

D. Boiling point depends only on the position of the OH group

Q27 Why do alcohols boil higher than alkanes and ethers of similar molecular weight?

	mwt	bp °C
CH_3OH	32	65
CH_3CH_3	30	−89
$CH_3CH_2CH_2OH$	60	97
$CH_3OCH_2CH_3$	60	11

A. Alcohols form hydrogen bonds between their molecules

B. Ethers and alkanes form stronger hydrogen bonds

C. Alcohols are ionic compounds with strong charges

D. Alcohols have a much higher molecular weight

Q28 Which statement about the water solubility of alcohols is correct?

A. Low-Mwt alcohols are insoluble; solubility rises as the hydrocarbon part grows

B. Low-Mwt alcohols are soluble; solubility falls as the hydrocarbon part grows

C. Solubility in water rises as the hydrocarbon part grows

D. All alcohols, whatever their size, mix fully with water

Q29 Hydrogen bonding occurs when hydrogen is bonded to:

A. A metal atom (Na, K)

B. A halogen only (Cl, Br, I)

C. A carbon atom (C)

D. A strongly electronegative atom (N, O or F)

8.4 Preparation of Alcohols

Q30 Which is **NOT** a method of preparing alcohols?

A. Hydration of alkenes

B. Reduction of aldehydes and ketones

C. Nucleophilic substitution of alkyl halides

D. Oxidation of alcohols with PCC

Q31 Ethene + H_2O / H^+ gives:

A. CH_3CH_2OH

B. $HOCH_2CH_2OH$

C. CH_3OH

D. CH_3CHO

Q32 $CH_3CH_2CH_2Br$ + NaOH gives:

A. $CH_3CH_2CH_2OH$, by electrophilic addition

B. $CH_3CH_2CH_2OH$, by nucleophilic substitution

C. $CH_3CH_2CH_2ONa$, by nucleophilic substitution

D. $CH_3CH_2CH_2ONa$, by electrophilic addition

Q33 Catalytic hydrogenation (H_2, Ni, pressure) of a ketone gives:

A. A tertiary alcohol

B. A secondary alcohol

C. A primary alcohol

D. A carboxylic acid

Q34 In the doctor's table, acetone (CH_3COCH_3) is reduced with $NaBH_4$ or $LiAlH_4$. The product is:

A. $(CH_3)_3COH$ (a 3° alcohol)

B. CH_3COOH (a carboxylic acid)

C. $CH_3CH(OH)CH_3$ (a 2° alcohol)

D. $CH_3CH_2CH_2OH$ (a 1° alcohol)

Q35 In the doctor's table, the aldehyde RCH_2CHO is reduced with $NaBH_4$ or $LiAlH_4$. The product is:

A. RCH_2CH_2OH (a 1° alcohol)

B. $RCH_2CH(OH)CH_3$ (a 2° alcohol)

C. RCH_2COOH (a carboxylic acid)

D. $RCH{=}CH_2$ (an alkene)

Q36 $NaBH_4$ and $LiAlH_4$ reduce aldehydes and ketones because they act as:

A. Free-radical sources

B. Oxidizing agents

C. Proton (H^+) donors

D. Hydride ($H:^-$) donors

Q37 A Grignard reagent (RMgX) reacts with formaldehyde, then H_2O/H^+, to give:

A. A secondary alcohol

B. A primary alcohol

C. A tertiary alcohol

D. A ketone

Q38 A Grignard reagent reacts with an aldehyde other than formaldehyde (HCOR′), then H_2O/H^+, to give:

A. A primary alcohol

B. A tertiary alcohol

C. A carboxylic acid

D. A secondary alcohol

Q39 A Grignard reagent reacts with a ketone (R″COR′), then H_2O/H^+, to give:

A. A tertiary alcohol

B. A primary alcohol

C. A secondary alcohol

D. An aldehyde

Q40 Grignard reagents are prepared from:

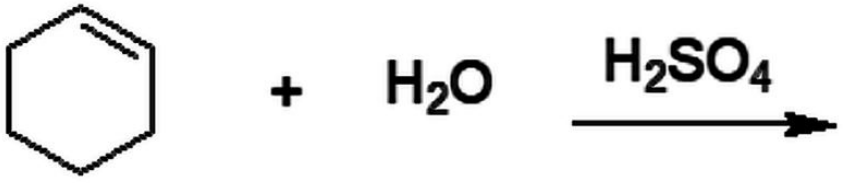
A. An alkene + H_2O with an acid catalyst

B. A ketone + $NaBH_4$ in water

C. An alkyl or aryl halide + Mg in diethyl ether

D. An alcohol + Na metal in dry diethyl ether

Q41 Complete: cyclohexene + H_2O / H_2SO_4 →



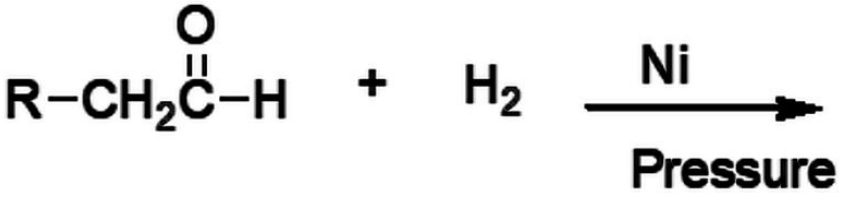
A. Cyclohexane-1,2-diol

B. Cyclohexanone

C. Cyclohexanol

D. Cyclohexane

Q42 Complete: $RCH_2CHO + H_2$ (Ni, pressure) →



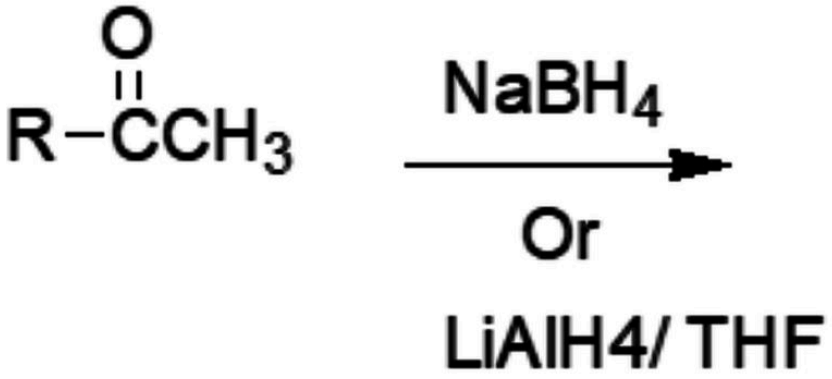
A. $RCH_2CH(OH)CH_3$

B. $RCH{=}CH_2$

C. RCH_2COOH

D. RCH_2CH_2OH

Q43 Complete: $RCOCH_3 + NaBH_4$ (or $LiAlH_4$/THF) →



A. RCH_2CH_3

B. $RCH(OH)CH_3$

C. RCOOH

D. RCH_2CH_2OH

8.5 Reactions of Alcohols

Q44 The reaction sites in an alcohol are:

A. One polar C–O bond and two non-bonding electron pairs on O only

B. Two polar bonds (C–O and O–H) and two non-bonding electron pairs on O

C. Only the C–H bonds of the alkyl group attached to O

D. A polar C=O bond, the O–H bond and one lone pair on O

Q45 Alcohols react with sodium metal to give:

A. An alkene (C=C) + NaOH

B. Sodium hydroxide (NaOH) + RH

C. Sodium alkoxide (RONa) + H_2

D. Alkylsodium (RNa) + H_2O

Q46 By which mechanism do alcohols react with hydrogen halides to produce alkyl halides?

A. Nucleophilic substitution

B. Elimination

C. Electrophilic substitution

D. Addition

Q47 Which statement about SN1 vs SN2 for alcohols + HX is correct?

A. SN1: 1 step, 2° and 3° alcohols; SN2: 2 steps, methyl and 1° alcohols, carbocation

B. SN1: 1 step, methyl and 1° alcohols; SN2: 2 steps, 2° and 3° alcohols, carbocation

C. SN1: 2 steps, methyl and 1° alcohols, carbocation; SN2: 1 step, 2° and 3° alcohols

D. SN1: 2 steps, 2° and 3° alcohols, carbocation; SN2: 1 step, methyl and 1° alcohols

Q48 The Lucas test (HCl in $ZnCl_2$) distinguishes 1°, 2° and 3° alcohols because:

A. Tertiary alcohols do not react with the reagent

B. They give different colours with $ZnCl_2$

C. They react at different rates, depending on carbocation stability

D. Only primary alcohols react, as they form the most stable carbocation

Q49 Besides HX, which reagents convert alcohols into alkyl halides?

A. $SOCl_2$ and PX_3

B. H_2SO_4 and H_3PO_4

C. $NaBH_4$ and $LiAlH_4$

D. PCC and CrO_3

Q50 By which mechanism do alcohols react with H_2SO_4 to produce alkenes (dehydration)?

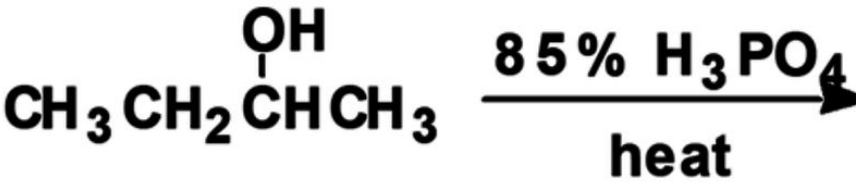
A. Addition reaction

B. Nucleophilic substitution

C. Electrophilic substitution

D. Elimination reaction

Q51 Dehydration of the alcohol shown (85% H_3PO_4, heat). What is the major product?



A. Butane

B. 2-Butene

C. 1-Butene

D. 2-Butanone

Q52 Oxidation of alcohols with Jones' reagent (CrO_3 in H_2SO_4) distinguishes them because:

A. 1° give aldehydes, 2° do not react, 3° give ketones

B. 1° do not react, 2° give ketones, 3° give carboxylic acids

C. 1° give carboxylic acids, 2° give ketones, 3° do not react

D. 1° give ketones, 2° give carboxylic acids, 3° do not react

Q53 Oxidizing a primary alcohol with PCC gives:

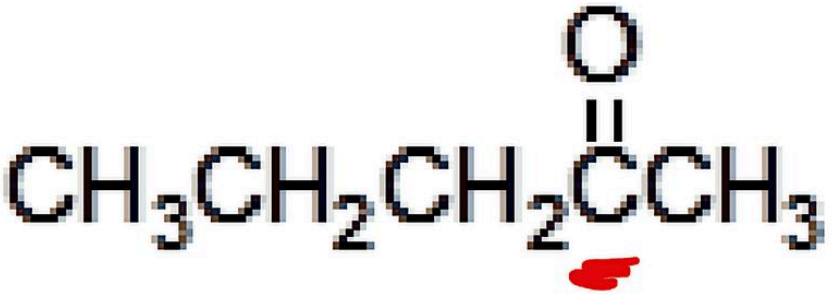
A. An aldehyde

B. No reaction

C. A ketone

D. A carboxylic acid

Q54 Which alcohol and oxidizing agent give $CH_3CH_2CH_2COCH_3$ (2-pentanone)?



A. 1-Pentanol + PCC

B. 1-Pentanol + CrO_3

C. 2-Methyl-2-butanol + CrO_3

D. 2-Pentanol + CrO_3

Q55 Which alcohol and reagent give $CH_3CH_2CH_2CH_2CHO$ (pentanal)?

A. Pentanoic acid + PCC

B. 1-Pentanol + PCC

C. 2-Pentanol + PCC

D. 1-Pentanol + CrO_3/H^+

Q56 Which alcohol and oxidizing agent give $CH_3CH_2CH_2CH_2COOH$ (pentanoic acid)?

A. 2-Pentanol + PCC

B. 1-Pentanol + CrO_3

C. 2-Pentanol + CrO_3

D. 1-Pentanol + PCC

Q57 A secondary alcohol (R_2CHOH) is oxidized with PCC or CrO_3/H^+. The product is:

A. A carboxylic acid

B. An aldehyde

C. A ketone

D. No reaction

Q58 Alcohols react with carboxylic acids (H_2SO_4 catalyst) to form:

A. Alkenes + water

B. Ketones + water

C. Ethers + water

D. Esters + water

Q59 Complete: 2-propanol + HBr → (mechanism?)

A. 2-Bromopropane + H_2O, SN1

B. 1-Bromopropane + H_2O, SN1

C. 2-Bromopropane + H_2O, SN2

D. 1-Bromopropane + H_2O, SN2

Q60 Complete: $CH_3CH_2CH_2OH + SOCl_2$ →

A. $CH_3CH{=}CH_2 + SO_2 + HCl$

B. $CH_3CH_2CH_2SO_2Cl + HCl$

C. $CH_3CH_2CHO + SO_2 + HCl$

D. $CH_3CH_2CH_2Cl + SO_2 + HCl$

Q61 Complete: 3 $CH_3CH(OH)CH_2CH_3$ + PBr_3 →

A. 3 $CH_3CH{=}CHCH_3$ + H_3PO_3

B. 3 $CH_3CHBrCH_2CH_3$ + H_3PO_3

C. 3 $CH_3CH_2CH_2CH_2Br$ + H_3PO_3

D. 3 $CH_3COCH_2CH_3$ + H_3PO_3

Q62 Complete: $CH_3CH(OH)CH(CH_3)CH_3$ + CrO_3/H^+ →

A. $CH_3COCH(CH_3)_2$

B. $(CH_3)_2C{=}CHCH_3$

C. $(CH_3)_2CHCH_2COOH$

D. No reaction

Q63 Complete: $CH_3(CH_2)_{10}CH_2OH$ + PCC →

A. $CH_3(CH_2)_9CH{=}CH_2$

B. No reaction

C. $CH_3(CH_2)_{10}CHO$

D. $CH_3(CH_2)_{10}COOH$

Q64 Complete: $CH_3CH(OH)CH_2CH_3$ + Na →

A. $CH_3CH{=}CHCH_3$

B. $CH_3CH(Na)CH_2CH_3$

C. $CH_3COCH_2CH_3$

D. $CH_3CH(ONa)CH_2CH_3$

Q65 Complete: $CH_3CH(OH)CH_2CH_3$ + H_2SO_4 →. Which alkene is the **minor** product?

A. $CH_3CH{=}CHCH_3$

B. $CH_2{=}CHCH_2CH_3$

C. $CH_2{=}C(CH_3)_2$

D. $CH_3CH_2CH_2CH_3$

Q66 Complete: $CH_3CH(OH)CH_2CH_3$ + HCl, $ZnCl_2$ →

A. $CH_3CH(Cl)CH_2CH_3$

B. $CH_3COCH_2CH_3$

C. $CH_3CH{=}CHCH_3$

D. $CH_3CH_2CH_2CH_2Cl$

Q67 Complete: $CH_3CH(OH)CH_2CH_3$ + CrO_3/H^+ →

A. $CH_3CH_2CH_2COOH$

B. $CH_3CH_2CH_2CHO$

C. $CH_3COCH_2CH_3$

D. No reaction

Q68 Dehydration of 3-methyl-2-butanol with H_2SO_4 gives mainly:

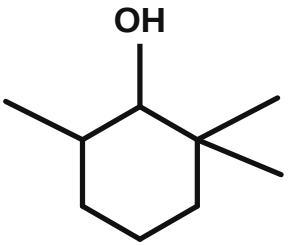
A. $(CH_3)_2CHCH{=}CH_2$

B. $(CH_3)_2CHCH_2CH_3$

C. $CH_3COCH(CH_3)_2$

D. $(CH_3)_2C{=}CHCH_3$

Q69 What is the major product of dehydration of this alcohol with H_2SO_4?



A. 1,1,2-Trimethylcyclohexane + H_2O

B. 1,3,3-Trimethylcyclohexene + H_2O

C. 2,2,6-Trimethylcyclohexanone + H_2O

D. No reaction; the alcohol is recovered

Answers · Lecture 8

Q1 B	**Q15** A	**Q29** D	**Q43** B	**Q57** C
Q2 C	**Q16** D	**Q30** D	**Q44** B	**Q58** D
Q3 A	**Q17** B	**Q31** A	**Q45** C	**Q59** A
Q4 D	**Q18** C	**Q32** B	**Q46** A	**Q60** D
Q5 D	**Q19** A	**Q33** B	**Q47** D	**Q61** B
Q6 A	**Q20** B	**Q34** C	**Q48** C	**Q62** A
Q7 C	**Q21** A	**Q35** A	**Q49** A	**Q63** C
Q8 B	**Q22** D	**Q36** D	**Q50** D	**Q64** D
Q9 A	**Q23** C	**Q37** B	**Q51** B	**Q65** B
Q10 B	**Q24** C	**Q38** D	**Q52** C	**Q66** A
Q11 C	**Q25** A	**Q39** A	**Q53** A	**Q67** C
Q12 D	**Q26** C	**Q40** C	**Q54** D	**Q68** D
Q13 C	**Q27** A	**Q41** C	**Q55** B	**Q69** B
Q14 B	**Q28** B	**Q42** D	**Q56** B	

9 Amines

In this lecture

1. Structure & Classification of Amines 7 questions
2. Nomenclature of Amines 8 questions
3. Physical Properties of Amines 4 questions
4. Basicity of Amines 8 questions
5. Preparation of Amines 5 questions
6. Reactions of Amines 7 questions
7. Aromatic Diazonium Compounds 16 questions

9.1 Structure & Classification of Amines

Q1 Amines are:

A. Oxygen-containing organic acids derived from water

B. Nitrogen-containing organic bases derived from ammonia

C. Nitrogen-containing organic acids derived from nitric acid

D. Carbon–nitrogen compounds that contain a C=O group

Q2 $(CH_3)_4N^+ Cl^-$ (R_4N^+) is a:

A. 4° alkylammonium salt

B. 4° amine

C. 1° alkylammonium salt

D. 3° alkylammonium salt

Q3 Amines are classified as 1°, 2° or 3° according to:

A. The number of hydrogen atoms on the carbon next to N

B. The type of carbon (1°, 2°, 3°) that carries the N

C. The number of nitrogen atoms in the molecule

D. The number of carbon groups bonded to the nitrogen

Q4 Classify $(CH_3)_3CCH_2NH_2$:

$$(CH_3)_3CCH_2NH_2$$

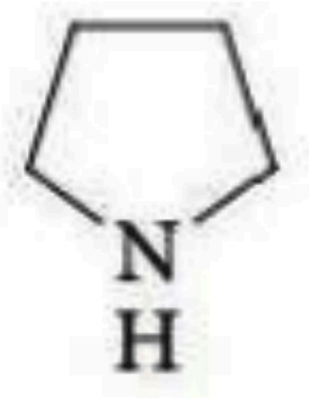
A. Quaternary

B. Secondary (2°)

C. Primary (1°)

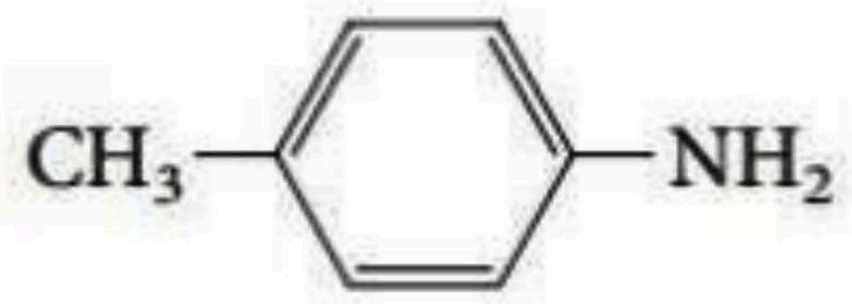
D. Tertiary (3°)

Q5 Classify this amine (pyrrolidine):



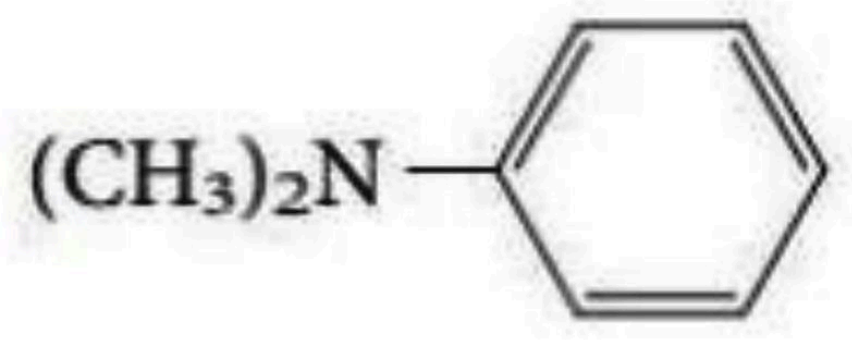
A. Primary (1°)
B. Secondary (2°)
C. Tertiary (3°)
D. Quaternary

Q6 Classify this amine:



A. Tertiary (3°)
B. Secondary (2°)
C. Quaternary
D. Primary (1°)

Q7 Classify this amine:



A. Quaternary
B. Primary (1°)
C. Tertiary (3°)
D. Secondary (2°)

9.2 Nomenclature of Amines

Q8 In common nomenclature, simple amines are named as:

A. Alkylamines

B. Alkanols

C. Alkanamides

D. Alkylammonium salts

Q9 In systematic (IUPAC) naming of amines, how is the chain numbered?

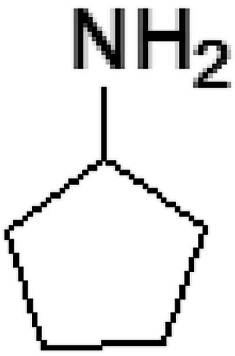
A. So that the alkyl substituents get the lowest numbers

B. From whichever end is nearer to the longest branch

C. So that the double bond always gets number 1

D. So that the carbon containing the amine group gets the lowest number

Q10 What is the name of this compound?



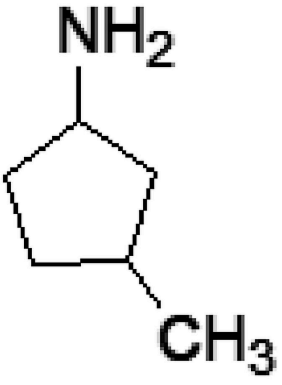
A. Aminocyclohexane

B. Cyclopentanamine

C. Cyclopentylamide

D. Pentanamine

Q11 What is the name of this compound?



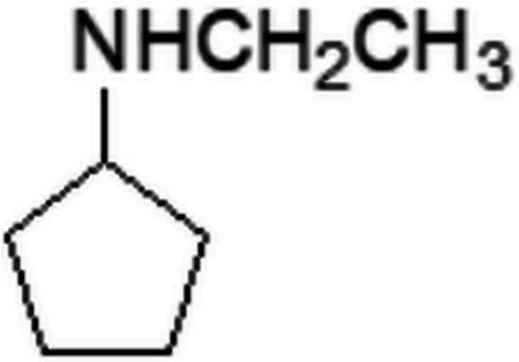
A. 4-Methylcyclopentanamine

B. 1-Methylcyclopentanamine

C. 3-Methylcyclopentanamine

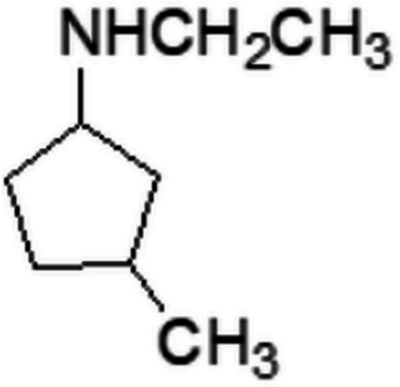
D. 2-Methylcyclopentanamine

Q12 What is the name of this compound?



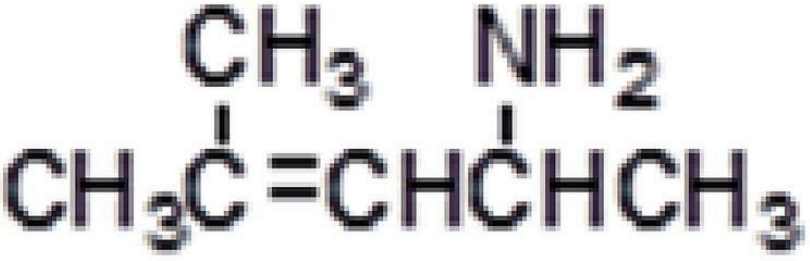
A. N-Ethylcyclopentanamine
B. Ethylcyclopentylamide
C. 2-Ethylcyclopentanamine
D. N-Methylcyclopentanamine

Q13 What is the name of this compound?



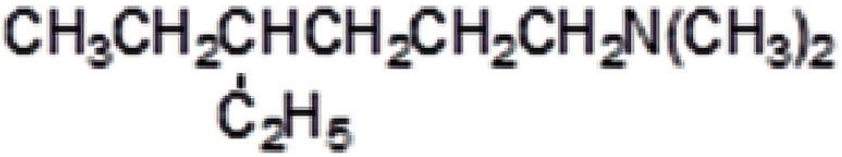
A. 1-Ethyl-3-methylcyclopentane
B. N-Ethyl-3-methylcyclopentanamine
C. 3-Ethyl-N-methylcyclopentanamine
D. N-Ethyl-2-methylcyclopentanamine

Q14 What is the IUPAC name of this compound?



A. 4-Methylpent-2-en-3-amine
B. 2-Methylpent-2-en-4-amine
C. 4-Methylpent-3-en-2-amine
D. 2-Aminopent-3-ene

Q15 What is the IUPAC name of this compound?



A. 4-Ethyl-N,N-dimethylhexan-1-amine

B. 3-Ethyl-N,N-dimethylhexan-6-amine

C. 4-Propyl-N,N-dimethylbutanamine

D. N,N-Dimethyl-4-ethylheptan-1-amine

9.3 Physical Properties of Amines

Q16 What can be concluded from the table?

Name	BP (°C)	Water sol. (mg/L)
Propane $CH_3CH_2CH_3$	−42	40
Ethanamine $CH_3CH_2NH_2$	17	Miscible
Ethanol CH_3CH_2OH	78	Miscible

A. Alcohol < amine < alkane in bp; small amines are insoluble

B. Alkane < amine < alcohol in bp; small amines are water soluble

C. Amine < alkane < alcohol in bp; small amines are insoluble

D. Alkane < alcohol < amine in bp; small amines are water soluble

Q17 Why do alcohols boil higher than amines of similar molecular weight?

A. Alcohols always have much higher molecular weights

B. Amines cannot form any hydrogen bonds with each other

C. O is more electronegative than N, so O–H hydrogen bonds are stronger

D. N is more electronegative than O, so N–H hydrogen bonds are weaker

Q18 Rank in order of **increasing** boiling point: $H_2NCH_2CH_2NH_2$, $(CH_3)_3N$, $CH_3NHCH_2CH_3$, $CH_3CH_2CH_2NH_2$

A. $H_2NCH_2CH_2NH_2 < CH_3CH_2CH_2NH_2 < CH_3NHCH_2CH_3 < (CH_3)_3N$

B. $CH_3NHCH_2CH_3 < (CH_3)_3N < H_2NCH_2CH_2NH_2 < CH_3CH_2CH_2NH_2$

C. All have the same boiling point

D. $(CH_3)_3N < CH_3NHCH_2CH_3 < CH_3CH_2CH_2NH_2 < H_2NCH_2CH_2NH_2$

Q19 Tertiary amines cannot form hydrogen bonds with each other.

True

False

9.4 Basicity of Amines

Q20 Amines are Lewis bases because:

A. The nitrogen has a non-bonding electron pair that it can share with H^+

B. The nitrogen has a positive charge that attracts electron pairs

C. The N–H bond breaks easily, so amines release H^+

D. The nitrogen has an empty orbital that accepts an electron pair

Q21 Which order of basicity is correct?

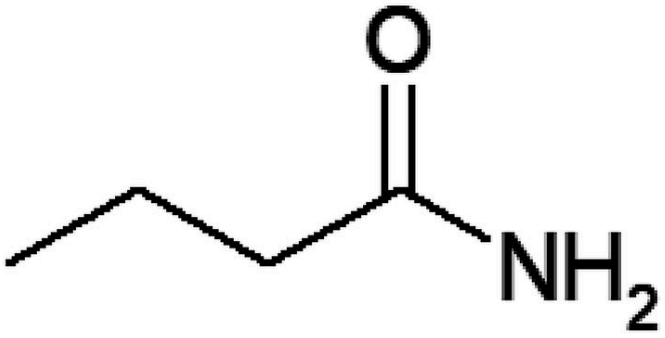
A. $NH_3 > CH_3NH_2 > (CH_3)_2NH$

B. $(CH_3)_2NH > CH_3NH_2 > NH_3$

C. $CH_3NH_2 > NH_3 > (CH_3)_2NH$

D. All are equally basic

Q22 Why is an amide (e.g. butanamide) much less basic than ammonia?



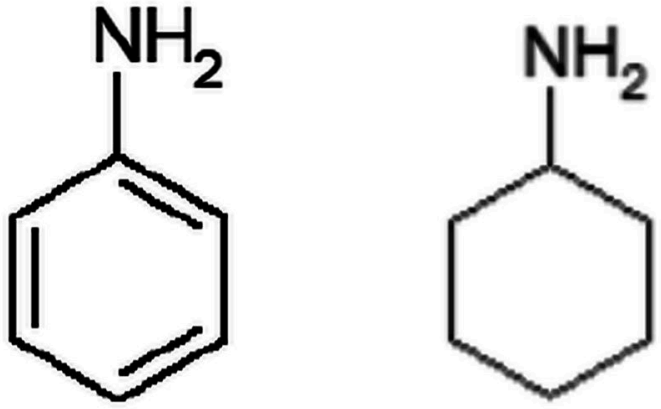
A. The C=O makes the amide ionic, so it cannot accept H^+

B. The N of an amide has no non-bonding electron pair at all

C. The alkyl chain (electron-releasing) pushes electrons away from N

D. The C=O (electron-withdrawing) delocalizes the N lone pair by resonance

Q23 Which is more basic, aniline (left) or cyclohexylamine (right)?



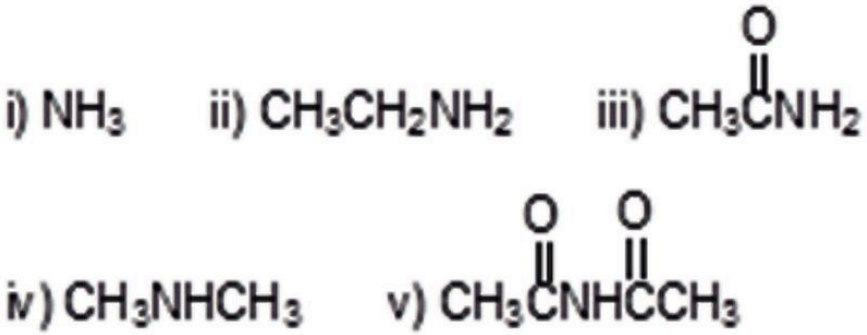
A. Neither is basic

B. Equal basicity

C. Cyclohexylamine

D. Aniline

Q24 Rank in order of **increasing** basicity:



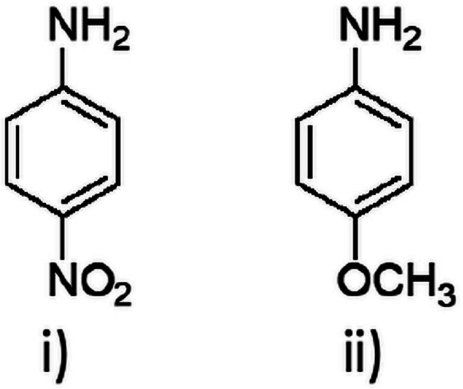
A. iii < v < i < iv < ii

B. v < iii < i < ii < iv

C. iv < ii < i < iii < v

D. i < ii < iii < iv < v

Q25 Rank in order of **increasing** basicity:



A. i = ii

B. ii < i

C. Neither is basic

D. i < ii

Q26 Why do substituents at the ortho and para positions of aniline have a larger effect on its basicity?

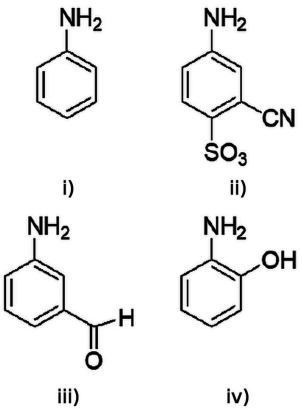
A. Ortho and para groups are always electron-releasing

B. Groups at ortho and para positions remove the lone pair from N completely

C. Resonance has special significance at the ortho and para positions

D. Inductive effects act only at the ortho and para positions

Q27 Rank in order of **increasing** basicity:



A. ii < iii < i < iv

B. iv < i < iii < ii

C. i < ii < iii < iv

D. iii < ii < iv < i

9.5 Preparation of Amines

Q28 Nitrobenzene is converted to aniline by:

A. Reduction with 3 H_2/Ni

B. Reaction with $NaNO_2$/HCl

C. Hydration with H_2O/H^+

D. Oxidation with $KMnO_4$

Q29 Butanamide ($CH_3CH_2CH_2CONH_2$) is reduced with $LiAlH_4$, then H_3O^+. What happens to the C=O group?

A. It is lost with one carbon, giving $CH_3CH_2CH_2NH_2$

B. It becomes CH=N, giving $CH_3CH_2CH_2CH{=}NH$

C. It becomes CH_2, giving $CH_3CH_2CH_2CH_2NH_2$

D. It becomes CH–OH, giving $CH_3CH_2CH_2CH(OH)NH_2$

Q30 Reduction of a nitrile (R–C≡N) with 2 H_2/Ni gives:

A. A secondary amine, $(RCH_2)_2NH$

B. A primary amine, $R–CH_2–NH_2$

C. A tertiary amine, $(RCH_2)_3N$

D. An amide, $R–CO–NH_2$

Q31 Why does alkylation of ammonia with alkyl halides give a mixture of products?

A. The alkyl halide is eliminated to an alkene before NaOH is added

B. The NaOH reacts with R–X to form alcohols instead of amines

C. Ammonia does not react with alkyl halides, so it stays unchanged

D. Each new amine is also a nucleophile, so it reacts again with the alkyl halide

Q32 Reductive amination of an aldehyde/ketone with a primary amine ($NaBH_3CN$) gives:

A. A primary amine

B. A secondary amine

C. An amide

D. A nitro compound

9.6 Reactions of Amines

Q33 Most reactions of amines are due to the non-bonding electron pair on N, which makes amines:

A. Acidic and nucleophilic

B. Acidic and electrophilic

C. Basic and electrophilic

D. Basic and nucleophilic

Q34 CH_3NH_2 + HCl gives:

A. $CH_3NHCl + H_2$

B. $CH_3Cl + NH_3$

C. $CH_3NH_3^+Cl^-$

D. No reaction

Q35 A useful application of forming amine salts with strong acids is:

A. Separating amines from non-basic organic compounds

B. Making amines insoluble in water so they precipitate

C. Converting amines into alkyl halides

D. Oxidizing amines to nitro compounds

Q36 Amines react with acid chlorides (RCOCl). Which statement is correct?

A. 1° amines give 3° amides, 2° give 2° amides, 3° do not react

B. 1° amines give 2° amides, 2° give 3° amides, 3° do not react

C. 1° amines do not react, 2° give 2° amides, 3° give 3° amides

D. 1° amines give 1° amides, 2° give 2° amides, 3° give 3° amides

Q37 Alkylation of amines with alkyl halides occurs by:

A. Free-radical substitution

B. Elimination (E2)

C. SN2 nucleophilic substitution

D. SN1 nucleophilic substitution (via a carbocation)

Q38 Complete: $(CH_3)_3NH^+Cl^- + CH_3NH_2 \rightarrow$

A. No reaction

B. $(CH_3)_3N + CH_3NH_3^+Cl^-$

C. $(CH_3)_3N + CH_3Cl + NH_3$

D. $(CH_3)_4N^+Cl^- + NH_3$

Q39 Complete: CH_3CONH_2 + $LiAlH_4$, then H_2O →

A. $CH_3CH_2NH_2$

B. CH_3COOH

C. CH_3CH_2OH

D. CH_3CN

9.7 Aromatic Diazonium Compounds

Q40 Aromatic diazonium salts are prepared from:

A. 3° aromatic amines + nitrous acid (HNO_2)

B. Aryl halides + sodium nitrite ($NaNO_2$)

C. Phenols + ammonia (NH_3)

D. 1° aromatic amines + nitrous acid (HNO_2)

Q41 The process that forms an aryl diazonium salt is called:

A. Reductive amination

B. Azo coupling

C. Diazotization

D. Acylation

Q42 How many important resonance forms does the aryl diazonium ion ($Ar–N_2^+$) have?

A. One

B. Four

C. Three

D. Two

Q43 Which resonance form of the aryl diazonium ion is important in the **substitution** (replacement) reactions?

A. $Ar–N=N^+$

B. $Ar–N^+\equiv N$

C. Ar–N=N–Ar′

D. $Ar–NH_2$

Q44 Which resonance form and mechanism are involved in the **coupling** reaction of aryl diazonium salts?

A. $Ar–N=N^+$ form; electrophilic aromatic substitution

B. $Ar–N=N^+$ form; nucleophilic substitution (SN2)

C. $Ar–N^+\equiv N$ form; electrophilic aromatic substitution

D. $Ar–N^+\equiv N$ form; free-radical substitution

Q45 In substitution (replacement) reactions of aryl diazonium salts:

A. N_2 is evolved and the ring is reduced

B. N_2 is retained and the ring is reduced

C. N_2 is retained and a nucleophile adds to the ring

D. N_2 is evolved and replaced by a nucleophile

Q46 Which reagent replaces the diazonium group by H?

A. HBF_4, heat

B. H_3O^+

C. H_3PO_2

D. KI

Q47 Benzenediazonium chloride + KI gives:

A. Iodobenzene + N_2

B. Aniline + I_2

C. Phenol + N_2

D. Benzene + N_2

Q48 Benzenediazonium chloride + H_3O^+ gives:

A. Chlorobenzene

B. Phenol

C. Aniline

D. Benzene

Q49 Which reagent converts benzenediazonium chloride into benzonitrile (C_6H_5CN)?

A. HCl, Cu_2Cl_2

B. KCN, $Cu_2(CN)_2$

C. HBF_4, heat

D. H_3PO_2

Q50 Benzenediazonium chloride + HCl, Cu_2Cl_2 gives:

A. Benzonitrile + N_2

B. Benzene + N_2

C. Chlorobenzene + N_2

D. Phenol + N_2

Q51 Benzenediazonium chloride + HBr, Cu_2Br_2 gives:

A. Chlorobenzene + N_2

B. Benzene + N_2

C. Phenol + N_2

D. Bromobenzene + N_2

Q52 Which reagent converts benzenediazonium chloride into fluorobenzene?

A. HBF_4, heat

B. HCl, Cu_2Cl_2

C. KI

D. H_3O^+

Q53 In coupling reactions, diazonium ions react with which kind of aromatic ring?

A. Deactivated rings carrying an EWG (NO_2, CN, CHO)

B. Any benzene ring, whatever its substituents

C. Highly activated rings carrying an ERG (OH, NH_2, NHR, NR_2, OR)

D. Only aryl halides (Ar–Cl, Ar–Br, Ar–I)

Q54 Nitrobenzene → 1) H_2 → 2) $NaNO_2$, HCl, 0 °C → 3) H_3PO_2 gives:

A. Chlorobenzene

B. Aniline

C. Phenol

D. Benzene

Q55 p-Nitrobenzenediazonium chloride + phenol gives:

A. $HO–C_6H_4–N=N–C_6H_4–NO_2$

B. $O_2N–C_6H_4–O–C_6H_5 + N_2$

C. $O_2N–C_6H_4–OH + N_2$

D. $HO–C_6H_4–NH–NH–C_6H_4–NO_2$

Answers · Lecture 9

Q1 B	**Q12** A	**Q23** C	**Q34** C	**Q45** D
Q2 A	**Q13** B	**Q24** B	**Q35** A	**Q46** C
Q3 D	**Q14** C	**Q25** D	**Q36** B	**Q47** A
Q4 C	**Q15** A	**Q26** C	**Q37** C	**Q48** B
Q5 B	**Q16** B	**Q27** A	**Q38** B	**Q49** B
Q6 D	**Q17** C	**Q28** A	**Q39** A	**Q50** C
Q7 C	**Q18** D	**Q29** C	**Q40** D	**Q51** D
Q8 A	**Q19** True	**Q30** B	**Q41** C	**Q52** A
Q9 D	**Q20** A	**Q31** D	**Q42** D	**Q53** C
Q10 B	**Q21** B	**Q32** B	**Q43** B	**Q54** D
Q11 C	**Q22** D	**Q33** D	**Q44** A	**Q55** A